

ANALYZING MACHINE LEARNING AND DEEP LEARNING TECHNIQUES FOR BRAIN TUMOR CLASSIFICATION ACROSS MRI, CT AND PET

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ABSTRACT

The diagnosis of brain tumors continues to be among the most difficult and deadly issues in the field of clinical neuroscience. Their early and accurate diagnosis is vital for the suitable treatment and patient outcomes. Neuroimaging methods like MRI, CT, and PET play an important role in diagnosis; however, manual interpretation is time-consuming and subject to variability. The introduction of artificial intelligence, especially machine learning (ML) and deep learning (DL), has a great impact on the medical image processing field.

In this paper, a systematic comparison of ML and DL methods for brain tumor classification across MRI, CT, and PET imaging modalities is performed. The research employs public datasets from Kaggle, including preprocessing, feature extraction, and model evaluation using Support Vector Machines (SVM) and Convolutional Neural Networks (CNNs). The performance was measured using different metrics like accuracy, precision, recall, F1-score, and ROC-AUC.

This research includes a procedure that covers data preparation, model training, and cross-modality benchmarking, thus exposing performance patterns specific to the modalities used. The findings show that the methods have a high degree of accuracy, with hybrid ResNet18 + SVM reaching 100% on PET and 99.77% on MRI binary classification for instance. Besides, a prediabetes risk scoring model is suggested but reformed here for tumor risk assessment. The research draws attention to the fact that automated systems are capable of augmenting clinical workflows while at the same time covering the shortcomings in multi-modal evaluations.

Table of Contents

Abstract.....	ii
Acknowledgement.....	iii
List of Tables.....	vii
List of Figures.....	viii
List of Abbreviations.....	ix
CHAPTER 1: INTRODUCTION.....	1
1.1 Background.....	1
1.2 Context.....	1
1.3 Importance of the Study.....	2
1.4 Current Challenges in Brain Tumor Classification.....	3
1.4.1 Limitations of Specific Modalities.....	3
1.4.2 Quality and Standardization of Data.....	3
1.4.3 Imbalanced Classes.....	4
1.4.4 Validation and Generalization.....	4
1.5 Research Problem.....	4
1.6 Aim of the Study.....	5
1.7 Research Objectives.....	5
1.8 Rationale for the Study.....	6

CHAPTER 2: LITERATURE REVIEW.....	7
2.1 Introduction.....	7
2.2 The Clinical Role of Multi-Modal Neuroimaging.....	7
2.3 Evolution of Automated Classification: From Handcrafted Features to Deep Learning.....	9
2.4 Machine Learning and Deep Learning Approaches in Brain Tumour Detection.....	10
2.5 Public Datasets for Brain Tumour Classification.....	11
2.6 Challenges in ML/DL for Brain Tumour Classification.....	12
2.7 Research Gaps and Motivation for the Present Study.....	13
2.8 Summary.....	14
 CHAPTER 3: RESEARCH METHODOLOGY.....	 15
3.1 Introduction.....	15
3.2 Research Design.....	15
3.3 Data Sources.....	16
3.4 Data Preprocessing Pipeline.....	19
3.4.1 Dataset Evaluation/Preparation.....	20
3.4.2 Image Standardization and Augmentation.....	20
3.5 Feature Engineering and Model Development.....	23
3.5.1 Classical ML Pathway: Support Vector Machine (SVM).....	23
3.5.2 Deep Learning Pathway - Convolutional Neural Network.....	23
3.5.3 Hybrid Exploration.....	24

3.6 Experimental Procedure.....	24
3.7 Evaluation Metrics.....	25
3.8 Implementation Environment.....	26
3.9 Ethical Considerations.....	27
3.10 Limitations of the Methodology.....	27
 CHAPTER 4: EXPERIMENTAL RESULTS AND EVALUATION.....	28
4.1 Background.....	28
4.2 Dataset Description.....	29
4.3 Data Preprocessing and Feature Extraction.....	31
4.4 Model Training Process.....	32
4.5 Experimental Results.....	32
4.5.1 Overall Accuracy and F1-Score.....	32
4.5.2 Classification Report Analysis.....	33
4.5.3 Confusion Matrix Dynamics.....	34
4.6 Discussion of Results Compared to Past Research.....	35
4.7 Interactive Demonstration of Results.....	35
4.8 Limitations of the Experimental Design.....	37
4.9 Conclusion.....	38

CHAPTER 5: DISCUSSION.....	39
5.1 Introduction.....	39
5.2 Analysing Experimental Results Through the Lens of Medical Imaging Research.....	40
5.3 Influence of Data Set Characteristics on Model Behaviour.....	41
5.4 The relationship of the research results to clinical requirements and operational constraints.....	43
5.5 Implications for Medical Imaging Practice.....	44
5.6 Deployment Feasibility and Clinical Implications.....	46
5.7 Study Limitations.....	46
5.8 Conclusion.....	47
 CHAPTER 6: CONCLUSION.....	 49
6.1 Chapter Summary.....	49
6.2 Summary of Major Findings.....	49
6.3 Practical Contributions.....	51
6.4 Academic Contribution.....	52
6.5 Limitations.....	53
6.6 Future Research.....	54
6.7 Final Comments.....	55

REFERENCES AND BIBLIOGRAPHY.....	56
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APPENDICES.....	58
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Appendix A – Python Code (written in Jupyter Notebook) Used in the Study.....	58
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Appendix B – Additional Experimental Results.....	58
---	----

Appendix C – Hardware & Software Environment.....	60
---	----

Appendix D – Dataset Description.....	60
---------------------------------------	----

Appendix E – Ethical Considerations.....	61
--	----

Appendix F – Project Folder Structure.....	62
--	----

List of Tables

Table 4.1: Summary of Dataset Statistics for MRI, CT, and PET Modalities.....	29
---	----

Table 4.2: Distribution of Healthy vs. Tumor Cases and Class Imbalance Across Modalities.....	30
--	----

Table 4.3: Performance Metrics for MRI Binary Classification Using the Best Model (ResNet18 Features + SVM).....	33
---	----

Table B.1: Detailed Test Performance Metrics.....	58
---	----

Table B.2: Confusion Matrix Highlights (Selected Examples).....	59
---	----

Table B.3: Label Distribution Across Modalities.....	59
--	----

Table B.4: Feature Summary.....	25
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List of Figures

Figure 3.1: Sample MRI Images of types of Tumors, and Healthy Brain Tissue.....	17
Figure 3.2: Sample CT Images of types of Tumors, and Healthy Brain Tissue.....	18
Figure 3.3: Sample PET Images of types of Tumors, and Healthy Brain Tissue.....	19
Figure 3.4: Randomly Augmented MRI Images for Binary Data Augmentation.....	21
Figure 3.5: Randomly Augmented CT Images for Binary Data Augmentation.....	21
Figure 3.6: Randomly Augmented PET Images for Binary Data Augmentation.....	22
Figure 4.1: Confusion Matrices for Binary and Multiclass Tasks Across Modalities.....	34
Figure 4.2 : Web Application Interface for Brain Tumor Classification.....	36
Figure 4.3 : Example Prediction Result from the Web Application.....	36
Figure 4.4 : Accuracy Comparison Between Handcrafted + SVM and ResNet18 + SVM Across Modalities.....	37
Figure 5.1: ROC Curves for the Best Model (ResNet18 Features + SVM) Across All Modalities and Tasks.....	40

List of Abbreviations

AI – Artificial Intelligence

CAD – Computer-Aided Diagnosis

CNN – Convolutional Neural Network

CT – Computed Tomography

DL – Deep Learning

FDG – Fluorodeoxyglucose

FLAIR – Fluid-Attenuated Inversion Recovery

GLCM – Grey-Level Co-occurrence Matrix

GPU – Graphics Processing Unit

k-NN – k-Nearest Neighbours

ML – Machine Learning

MRI – Magnetic Resonance Imaging

PACS – Picture Archiving and Communication System

PET – Positron Emission Tomography

PII – Personally Identifiable Information

RBF – Radial Basis Function

RIS – Radiology Information System

ROC-AUC – Receiver Operating Characteristic – Area Under the Curve

RSNA – Radiological Society of North America

SUV – Standardized Uptake Value

SVM – Support Vector Machine

T1 – T1-weighted MRI

T2 – T2-weighted MRI

WHO – World Health Organization

XAI – Explainable Artificial Intelligence

XGBoost – Extreme Gradient Boosting

CHAPTER 1 : INTRODUCTION

1.1 Background

Brain tumors still represent one of the most difficult and deadly conditions in clinical neuroscience, where having an early and accurate diagnosis becomes the main factor in the treatment and hence, in the patient's recovery (Aamir et al., 2022). Nowadays, neuroimaging modalities or techniques such as MRI, CT, and PET are the most vital methods in clinical practice for diagnosing and managing brain tumors. Yet, the interpretation of these complex imaging datasets is still tedious and suffers from the variability in human judgment, observer variation, and, to a lesser extent, fatigue (Rasheed et al., 2023).

The birth of AI, especially machine learning (ML) and deep learning (DL), has changed the whole scenario of medical image analysis. Classical ML algorithms like SVM depend on the process of feature extraction where certain parameters such as the object's shape, texture, and intensity are defined and isolated manually to prepare them for training (Basthikodi et al., 2024). Their output may be good enough for some applications, but the limited nature of expert-defined features and so the accordingly limited model dependence, makes the models non-applicable in many diverse imaging modalities. On the other hand, deep learning models, particularly CNNs, free up the process of feature mining completely and do not need any prior expert input at all as they automatically learn hierarchies from raw pixel data, thus, potentially achieving higher accuracy in classification tasks (Xie et al., 2022).

Even with the rapid progress made, the current body of research has still been oriented mainly toward single-modality datasets, with MRI as the dominant imaging technique, which has resulted in a major drawback in algorithmic performance being evaluated systematically across modalities (Riahi et al., 2024). Each imaging modality is still complementary to the other in the sense that it can bring about the emergence of the other.

1.2 Context

Research into the classification of brain tumors currently showcases a variety of methods, but the lack of systematic comparisons among them is a downside. The majority of the studies conduct their evaluations on single-modality datasets that have been rigorously curated and thus the full range of clinical imaging variability is not represented. The dataset of the Kaggle "Brain MRI Images for Brain Tumour Detection" has turned out to be the principal source of many studies, but its restriction to MRI only does not reveal the performance of CT and PET (Aamir et al., 2022).

In academia, it has been proven that deep learning models based on CNN outperform all other methods in the field of tumor detection in MRI scans, but the question of their performance across CT and PET modalities remains unanswered(Xie et al., 2022). Traditional ML techniques still prove to be beneficial especially when paired with advanced feature extraction methods implying that both classical and modern ones should be put under comparative evaluation(Basthikodi et al., 2024).

Another major obstacle is the practical implementation gap between laboratory results and clinical practice. Different preprocessing pipelined, augmentation strategies, and evaluation metrics are often used by studies, thereby making it difficult to compare and limiting the development of evidence-based guidelines for clinical implementation (Rasheed et al., 2023).

1.3 Importance of the Study

The present study systematically compares machine learning and deep learning approaches across MRI, CT, and PET imaging modalities, thus filling critical gaps in the current literature and providing new insights on the subject. The significance of this study can be expressed in various ways.

- **For clinical practice:**

By performing a benchmark of the various modalities, the research gives the clinical decision-makers the basis for their choice of imaging technologies and automated processes in different clinical situations.

- **For research advancement:**

The study takes Riahi et al.'s (2024) identified issue with the scarcity of cross-modality evaluation as the starting point. The researchers' application of the same preprocessing and evaluation protocols over the three modalities has enabled them to make significant performance comparisons and help methodological standardization.

- **For technological development:**

Knowledge of the performance patterns specific to each modality should help in the development of future algorithms, resulting in a more reliable and flexible automated diagnostic system that can work flawlessly with all imaging technologies.

- **For healthcare efficiency:**

The deployment of automated classification systems will not only lessen the diagnostic workload but also reduce inter-observer errors, leading to more effective and uniform diagnostic workflows as outlined by Raza et al. (2022).

1.4 Current Challenges in Brain Tumor Classification

Even though there have been technological improvements, several challenges that have existed for a long time still make it difficult to classify brain tumors automatically:

1.4.1 Limitations of Specific Modalities

Every imaging technique has its own set of problems. MRI has the best contrast for soft tissues but is susceptible to motion artifacts and needs longer acquisition times. CT, on the other hand, has the rapidest imaging but the softest tissue contrast is the lowest. PET gives functional information but with less spatial resolution. The characteristics of different modalities can severely affect the performance of algorithms (Riahi et al., 2024).

1.4.2 Quality and Standardization of Data

Differences in image acquisition protocols, preprocessing methods, and annotation standards across different datasets are the main reasons for the difficulties in building robust models that generalize well across clinical settings (Rasheed et al., 2023).

1.4.3 Imbalanced Classes

Difficulties with imbalanced classes often occur in brain tumor datasets, where certain types of tumors are represented to a much lesser extent. This can affect the automated systems negatively by biasing them and also reducing their performance for rare tumor types (Asiri et al., 2024).

1.4.4 Validation and Generalization

It is a common scenario for many automated systems to exhibit high-performance levels when assessed using curated datasets, but the same systems get difficult to apply in real clinical scenarios where there is higher variability in image quality and patient characteristics (Mahesh et al., 2023).

1.5 Research Problem

Even though a lot of research has been done on the automated classification of brain tumors, there are still considerable gaps in the understanding of the performance of different computational methods over the entire range of clinically relevant imaging modalities. The majority of the studies done so far have been concentrated on one imaging modality, mainly MRI, and this has created a fragmented knowledge that does not represent the multimodal nature of clinical practice (Riahi et al., 2024). Such a situation reduces the practical usage of research outcomes and inhibits the formation of complete diagnostic systems that would be able to combine data from various imaging sources.

Besides that, different methods used in the studies make it even harder to compare the results. The differences in the preprocessing of images, metrics of evaluation, and the design of the experiment lead to the situation that one cannot be sure about the relative performance of the algorithms in the

different modalities (Rasheed et al., 2023). This inconsistency in the methods used is one of the major obstacles to the research progress in the area.

The interaction among the different imaging modalities between classical machine learning methods and modern deep learning ones has not been well explored yet. Deep learning has been very effective in many MRI-based studies, but the extent to which it outperforms other methods may be modality-dependent (Xie et al., 2022). Knowing the different performance trends due to the characteristics of the modalities is very important in the selection of the approach for different clinical scenarios.

1.6 Aim of the Study

The primary aim of this research is to conduct a comprehensive comparative analysis of machine learning and deep learning techniques for brain tumor classification across MRI, CT, and PET imaging modalities, with the objective of identifying modality-specific performance patterns and providing evidence-based recommendations for approach selection in different clinical contexts.

1.7 Research Objectives

In order to reach this goal, the research will conduct a series of specific objectives:

1. The first objective is to thoroughly investigate and prepare MRI, CT, and PET brain images for classification purposes, taking care to apply the same standard to all modalities to guarantee high-quality inputs for the classification analysis.
2. The second objective is to apply and test numerous supervised learning techniques, both classical and deep learning, consisting of SVM and CNN respectively, for the purpose of detecting and classifying brain tumors using medical images.
3. The third objective is to evaluate and compare the classification methods using a variety of performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to find the best method for precise brain tumor detection through different imaging systems.

4. The fourth objective is to recognize the main aspects that influence the effectiveness of the classification in different imaging modalities and to give suggestions for improvements in the clinical workflow for brain tumor detection.

1.8 Rationale for the Study

This study fills a significant void in the evaluation of automated brain tumor classification systems across different modalities. Up to now, research mainly concentrated on single imaging modalities, whereas in the clinic, information from MRI, CT, and PET scans is combined. The lack of systematic comparisons between these modalities is one of the main reasons why the use of artificial intelligence in neuro-oncology is still limited. By analyzing traditional machine learning (Support Vector Machines) and deep learning (Convolutional Neural Networks) methods on all three modalities, this work reveals the insights on which clinical decision-making and diagnostic workflow improvement can be based.

Moreover, the research facilitates the establishment of a standard methodology in the field of medical AI through the application of consistent evaluation protocols across different modalities. Such a strategy sets up the performance benchmarks that are reliable and incorporate both the accuracy and computational requirements, thus making the findings particularly useful for hospitals with limited resources. The research has shown that AI can be better integrated into healthcare if sufficient, multi-modal evaluations are provided which mirror the actual diagnostic practice.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Brain tumor classification remains a very critical domain in the field of medical diagnostics. The timely and correct identification of a tumor could have a great influence on surgical planning, treatment efficacy, and patient survival. Clinically, different combinations of imaging techniques can be employed-major ones like MRI, CT, and PET-which are employed by radiologists to study the anatomic structure, tissue density, and metabolic activities. However, the interpretation of these images manually is marred by a number of factors like the subjective nature of expert opinions, fatigue from making patient diagnoses, and a continuous increase in the imaging data; these would all contribute to late decisions and inconsistencies. According to Aamir et al. (2022), in this regard, one of the major strategies that neuro-oncology and artificial intelligence researchers had to resort to was the development of automatic CAD systems supported by machine learning and deep learning technologies. These machines are used for objectively classifying brain tumors with consistency and swiftness to support clinical decision-making and improve diagnostic workflows.

The present chapter therefore provides a detailed review of state-of-the-art research for classifying brain tumors using ML and DL techniques in MRI, CT, and PET modalities. We begin by relating the medical setting, particularly the importance of every imaging modality in tumor diagnosis. The chapter follows the chronological development of these methods from early systems based on features to the current deep learning architectures. Further, we discuss the present state of ML and DL approaches in brain tumor classification, pointing out multi-modal data usage as an area of specific interest. The various public datasets available for research purposes are discussed, with special attention to those hosted on Kaggle .

2.2 The Clinical Role of Multi-Modal Neuroimaging

The treatment of brain tumors is a complicated procedure involving multiple tests, or 'imaging modalities.' Imaging modalities provide diagnostic information about brain tumors but cannot

individually provide a complete diagnosis. Therefore, various imaging modalities should always be used together to give the most accurate indication of the brain tumor.

The MRI (Magnetic Resonance Imaging) is the primary imaging modality used for brain tumors. An MRI provides excellent contrast between soft tissues and no ionising radiation is used. The MRI allows the radiologist to view in detail the tumour margins and amount of swelling (oedema) surrounding the tumor, as well as the involvement of important structures of the brain. Different MRI scans (T1 plus contrast, T1 without contrast, T2, and FLAIR) enable radiologists to identify different brain tumor subtypes and to evaluate the density and necrosis (i.e., dead cells) associated with the tumors. As a result, an MRI scan is used as the primary procedure for initial identification of the brain tumor, for surgical planning, and for post-treatment assessments (Riahi et al., 2024).

CT (Computed Tomography) is useful during emergency situations because it can be obtained very quickly. CT scans are excellent for evaluating bony structures, areas of calcification (e.g., in meningiomas or oligodendrogliomas), and recent bleeding. For patients who cannot undergo an MRI procedure (due to hardware or the high anxiety associated with being in the machine), CT scans are the imaging procedure used first. Although CT scans do not provide the same level of soft-tissue contrast as MRI scans, they still provide important structural and density information about the brain tumor (Rasheed et al., 2023).

Positron Emission Tomography (PET) is able to represent both functionality and metabolism through the use of radiolabelled compounds, most commonly Fluorodeoxyglucose (FDG). Tumour cells use elevated levels of glucose for metabolism, which allows the use of PET to distinguish between different grade tumours. The use of PET to help differentiate tumour recurrence from post-radiation necrosis and to help identify metastatic disease is a vital component of treatment planning and the assessment of treatment response because it provides an additional layer of assessment beyond an anatomical basis (Boehm et al., 2022).

The combination of these imaging modalities (generally referred to as the 'anatomy (MRI/CT) + Functionality (PET) approach') forms the basis of clinical excellence, however, developing automated solutions that use information from all three imaging modalities simultaneously is a major computational obstacle in delivering optimal imaging solutions.

2.3 Evolution of Automated Classification: From Handcrafted Features to Deep Learning

The evolving nature of the development of systems that automatically classify brain tumours corresponds to the changing nature of artificial intelligence, which is being developed in phases.

Prior to 2010, CAD systems used radiomics and classic machine learning methods in which experts first extracted feature sets from digital images by calculating features. These included first-order statistics (the average intensity and skewness) and second-order texture features that were calculated from a Grey-Level Co-occurrence Matrix (GLCM) (Haralick et al., 1973), as well as shape descriptors (sphericity, surface area) before training classifiers (such as Support Vector Machine (SVM), Random Forest, or k-Nearest Neighbour (k-NN)) based on these features.

CADs offered interpretability; however, their reported accuracy on the many clinical datasets was typically between 50% to 75% (Lambin et al., 2017), and they had limitations concerning both feature engineering and lack of understanding of the underlying features. Specifically, feature extraction is a labour-intensive process that requires professionals with domain expertise and not all necessary features have been defined sufficiently to represent the complexities and intricacies of tumour type identification across imaging protocols and scanners.

Deep learning, particularly Convolutional Neural Networks (CNNs), changed the entire game in the area after their discovery in the 2012 ImageNet competition (Krizhevsky et al., 2012). CNNs, through the successive application of various convolutional, pooling, and non-linear activation layers, get the pixel-based images to represent the high-level features. The original architectures like AlexNet, VGG, and later ResNet (He et al., 2016) and DenseNet were finely tuned for different medical images. With their unparalleled capability of detecting even the most intricate spatial patterns, these models achieved and held the top performance in image classification and segmentation tasks. Filtering and classifying brain tumours, CNNs applied to MRI data has always been getting accuracy rates of over 90% on standard datasets, much higher than the previous radiomic methods (Xie et al., 2022).

2.4 Machine Learning and Deep Learning Approaches in Brain Tumour Detection

Current research is utilizing a wide range of methods varying from simple and classical ML to complex and deep-learning architectures with each having its own merits and demerits.

Classical Machine Learning with Engineered Features: The classic ML models still hold their ground despite the DL wave, particularly in the midst of the limited data and processing power scenarios. The machines that work based on the Support Vector (SVMs) algorithm, along with the non-linear kernels (like Radial Basis Function) can very efficiently decide the boundaries in very high dimensional feature spaces. Works such as that of Basthikodi et al. (2024) have shown that SVMs where the textures and shapes are very meticulously selected can still reach a pretty strong performance of more than 85% (accuracy) on very well-structured MRI datasets. Random Forests and Gradient Boosting methods (e.g., XGBoost) are endowed with calculated feature importance and have the character of not being easily devoured by overfitting, thus, they are considered to be the right methods for heterogeneous data. Nonetheless, the apex of their performance is inherently connected to the properties of the input features.

Deep Learning Architectures: The DL models of 2D and 3D CNNs especially have taken the lead with these superior ones being a major part of the current research. The 2D CNNs deal with each single slice of the image, they are less demanding on the computation side and they are very much adapted to studying datasets that consist of 2D scans (which is already the case in public repositories). The 3D CNNs take in volumes of data and would capture the interslice contextual information that could bolster accuracy for volumetric classification but with a profound increase in the cost of computation. Transfer Learning which involves the fine-tuning of the CNNs that are pre-trained on huge natural image datum like ImageNet—has become a common practice to deal with the shortage of large, labeled medical datasets (Yosinski et al., 2014). This method makes it possible for the models to utilize generic visual feature detectors, achieving high performance even with limited medical training samples.

Hybrid and Multi-Modal Strategies: Based on the actual clinical situation of multi-modal imaging, the investigation of fusion methods has just started. Early fusion merges different modalities' raw and pre-processed images at the input level. Late fusion trains separate models for each modality and combines their predictions (e.g., through averaging or a meta-classifier). Late fusion is usually more practical as it allows for missing modalities. Nevertheless, MRI sequences or MRI with CT

are the predominant combinations in multi-modal studies; PET's inclusion is still rare because the datasets are not sufficient (Lipkova et al., 2023).

2.5 Public Datasets for Brain Tumour Classification

The crucial part that standardised, publicly accessible datasets play in benchmarking, as well as in making research progress, is undeniable. The present project which employs 2D image data, gets its pivotal support from Kaggle.

Among the most noteworthy resources is Kaggle "Brain MRI Images for Brain Tumour Detection," which has gradually become the most reliable public dataset for brain tumor classification based on 2D MRI. The dataset comprises around 3,000 T1-weighted contrast-enhanced MRI slices divided into four categories: glioma, meningioma, pituitary tumor, and tumor-free. Its merits are well-defined labeling, equal distribution of classes (for a medical dataset), and usability in an immediate manner for 2D CNN training. On the other hand, its drawbacks are the restriction to only one type of MRI sequence and the unavailability of corresponding CT or PET data, which inhibits a cross-modality study.

Kaggle Brain CT & PET Datasets: These repositories consist of brain CT and PET images. The datasets may include CT slices for hemorrhaging detection or PET scans for Alzheimer's disease; the main difference is that they do not have that much exposure as MRI datasets. Still, they can be repurposed, merged, and made ready for tumor classification tasks. One of the major challenges related to this matter is that these collections of CT and PET images are usually not associated with the Kaggle MRI dataset or each other. They are taken from different patient groups, have distinct labeling schemes (e.g., tumor versus no tumor, or different pathology altogether), and differ in resolution and quality. This situation reveals one of the biggest drawbacks in the public data landscape: the lack of a big, co-registered, and uniformly labeled multi-modal (MRI-CT-PET) brain tumor dataset. Consequently, researchers resort to the painstaking process of curation, preprocessing, and normalization so as to make comparative analysis possible across these independent modality-specific datasets.

2.6 Challenges in ML/DL for Brain Tumour Classification

The reliability and robustness of automated classification systems will continue to be challenged by the same old problems even if the results were already good.

- **Data Scarcity and Heterogeneity:**

It is hard to get large, high-quality, labelled medical datasets because of the privacy issues, the costs and the skills needed for annotation. Besides, the performance of the model gets severely reduced when applied to data from new institutions owing to domain shift that is caused by differences in scanner manufacturers, imaging protocols, and patient populations.

- **Class Imbalance:**

The different types of brain tumours are not equally prevalent in nature. For example, gliomas are more common than pituitary tumours. Models trained on imbalanced data become biased towards the majority class, showing high overall accuracy but poor sensitivity for rare tumour types which can be clinically critical.

- **Modality-Specific Technical Challenges:**

Every imaging modality has its own difficulties. CT images have lower soft-tissue contrast and are susceptible to beam-hardening artefacts. PET images are characterized by low spatial resolution and high noise levels, and their interpretation is highly dependent on standardised uptake values (SUVs) which need careful normalisation.

- **Computational Resource Demands:**

The most advanced 3D DL models are very demanding in terms of GPU memory and training time that can take days, thus making it difficult for researchers and clinicians without access to high-performance computing infrastructure to participate in the development of such models.

- **Interpretability and Clinical Trust:**

The "black-box" characteristic of complex DL models is one of the main factors that hinder their adoption in clinical practice. Radiologists must comprehend the rationale behind an AI's prediction in order to trust and act upon it. Despite the fact that explainable AI (XAI) methods such as Grad-

CAM are being developed, they are still not regarded as standard or entirely satisfactory for making high-stakes medical decisions.

2.7 Research Gaps and Motivation for the Present Study

The literature analysis shows overlapping gaps that will be directly tackled by this research:

- **Modality-Isolated Research :**

Most of the studies limit their focus only to one imaging modality which is mainly MRI. There is a big lack of systematic, comparative research that measures and sets the level of the same ML/DL algorithms across all three important modalities (MRI, CT, and PET) using an identical experimental framework.

- **Dataset-Driven Limitations :**

The lack of one common, multi-modal public dataset isolates the researchers and makes them rely on different, unpaired sources (such as the separate Kaggle datasets for MRI, CT, and PET). Very few researchers directly take up the issue of creating and testing models under this realistic constraint, which is similar to the "messy" data environment that is often found in clinical practice.

- **Underexplored Efficient Hybrid Models :**

While deep learning is the winner in performance, resource-efficient hybrid pipelines e.g., using a pre-trained CNN as a feature extractor followed by a classical SVM classifier have not been put through extensive benchmarking against end-to-end DL models across MRI, CT, and PET. This particular gap in the area of practical deployment is very important as computational efficiency is a top priority there.

- **The Lack of Attention to Cross-modal Generalization :**

There are not many studies that examine whether features or models acquired in one modality (e.g., a dataset-rich modality such as MRI) can help improve classification performance in another modality (e.g. a less-desired view of data (less-eriest in this example) from PET). We need more

research on the relationship between different modalities so that we can build data-efficient models.

To our knowledge, this study fills this void. By utilizing distinct Kaggle datasets of MRI, CT and PET scans, we will build, train and validate a classical machine learning (ML) framework (SVMs using the Deep Features) and a modern deep learning (DL) framework (2D Convolutional Neural Networks (CNNs)) on each of these modalities independently, leading to a clear empirical comparison of ML and DL models across all three neuro-oncology diagnostic domains, resulting in knowledge about creating versatile, practical computer-aided diagnostic (CAD) tools.

2.8 Summary

A comprehensive literature review was given in this chapter regarding the use of machine learning and deep learning for the classification of brain tumours. It pinpointed the significant clinical functions of MRI, CT, and PET and traced the journey of the field from radiomics to deep learning. The existing methodological techniques were evaluated; these comprise classical ML, deep CNNs, and hybrid methods. The review underscored the significance of public datasets, especially the ones on Kaggle, but it also referred to the major difficulty posed by the absence of integrated multi-modal data. Key technical and practical challenges—from data heterogeneity to the "black box" problem—were discussed. Lastly, specific research gaps were outlined, focusing on the lack of comparative multi-modal studies using disjoint datasets and the need for efficiency-aware model evaluation. These gaps are the basis for the present research, designed to provide a thorough, modality-wise analysis of classification techniques to usher the field toward more robust and clinically applicable diagnostic support systems.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

This Chapter elaborates on the methodological and systematic framework used in this Research to create a comparison of the effectiveness of Machine Learning (ML) and Deep Learning (DL) methods used in Brain Tumor Classification, for three different imaging methods, such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT), and Positron Emission Tomography (PET). In this study, a Quantitative Experimental Comparative Research design based on a Positivist research philosophy has been used to develop knowledge from objective measurements, the application of algorithms and the ability to replicate through analysis (Creswell & Creswell, 2018). Since the study's goal is to assess how the performance of models can be compared across different datasets and modalities, the methodology was developed to ensure that all data was prepared, all models were trained in the same way, and the evaluation of each model was based on standardized performance measures.

The Chapter provides an overview of the complete progression of the research from Data Acquisition through to final analysis, as well as a description of the Research Design, and a description of the Public Datasets obtained from Kaggle, followed by an explanation of the Multi-Stage Data Preprocessing and Augmentation Pipeline, the extraction of features from the Data Set, the Creation of a Model using both Classical and DL techniques, the Procedure for Training and Validating the Model, the Performance Metrics selected, the technology used for Implementation, as well as the Ethical Issues and Limitations in the Methodology. Justifications for all methodological decisions made were referenced back to current medical image analysis and machine learning research practices.

3.2 Research Design

The research design in this study is a comparative experiment that uses deductions to conduct and evaluate the performance of several algorithms over a number of different experimental conditions, such as SVMs compared to CNNs, and in a controlled, repeatable manner (Géron, 2022). The

research design is a quantitative methodology that uses metrics such as accuracy, precision, recall and F1-score to provide statistically valid conclusions about which of the algorithms is more effective.

One of the most important features of the design is that it has a parallel structure by modality. The processing of the MRI, CT and PET data, the training and evaluation of the model, are all performed separately for each of these modalities. By doing this, it will be possible to directly compare the performance of the SVM and CNN algorithms on the same dataset. After that, the results will be integrated across the modalities to get an overall trend and insights into how the modalities differ from each other. A clearly defined, repeatable research design reduces the potential for confounding variables between the different modalities and provides strong evidence that the differences in performance of the algorithms are due to either the type of algorithm or the unique characteristics of each modality and not due to inconsistencies in the research process.

3.3 Data Sources

For the analysis, the researchers parade publicly available, secondary medical imaging datasets procured through the Kaggle platform. The employment of ready-for-use, anonymised data dovetails with the project's low-risk ethical classification and supports replicability.

The major datasets used are:

- 1. Kaggle "Brain MRI Images for Brain Tumour Detection":** This dataset is the mainstay of MRI-based investigation and it contains roughly 3,000 T1-weighted contrast-enhanced axial brain MRI slices that have been divided into four categories: Glioma, Meningioma, Pituitary tumour, and No Tumour.

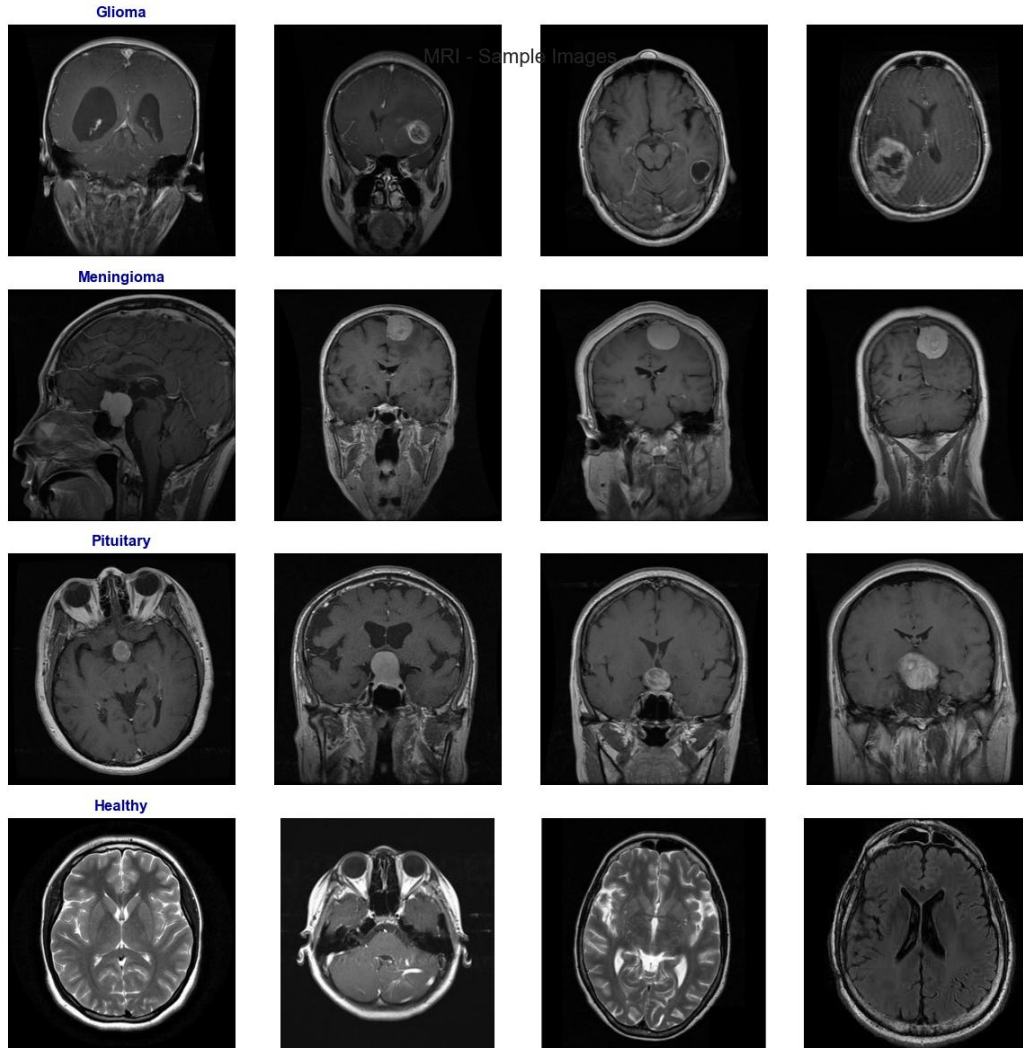


Figure 3.1: Sample MRI Images of types of Tumors, and Healthy Brain Tissue.

2. Kaggle Brain CT Images Dataset:

A set of brain CT scans that have been selected and adapted for this research. The images are sorted into two classes (Tumor vs. Healthy) for the sake of consistency with the major classification task, but it is also recognised that care must be taken in selection since the images come from a different group.

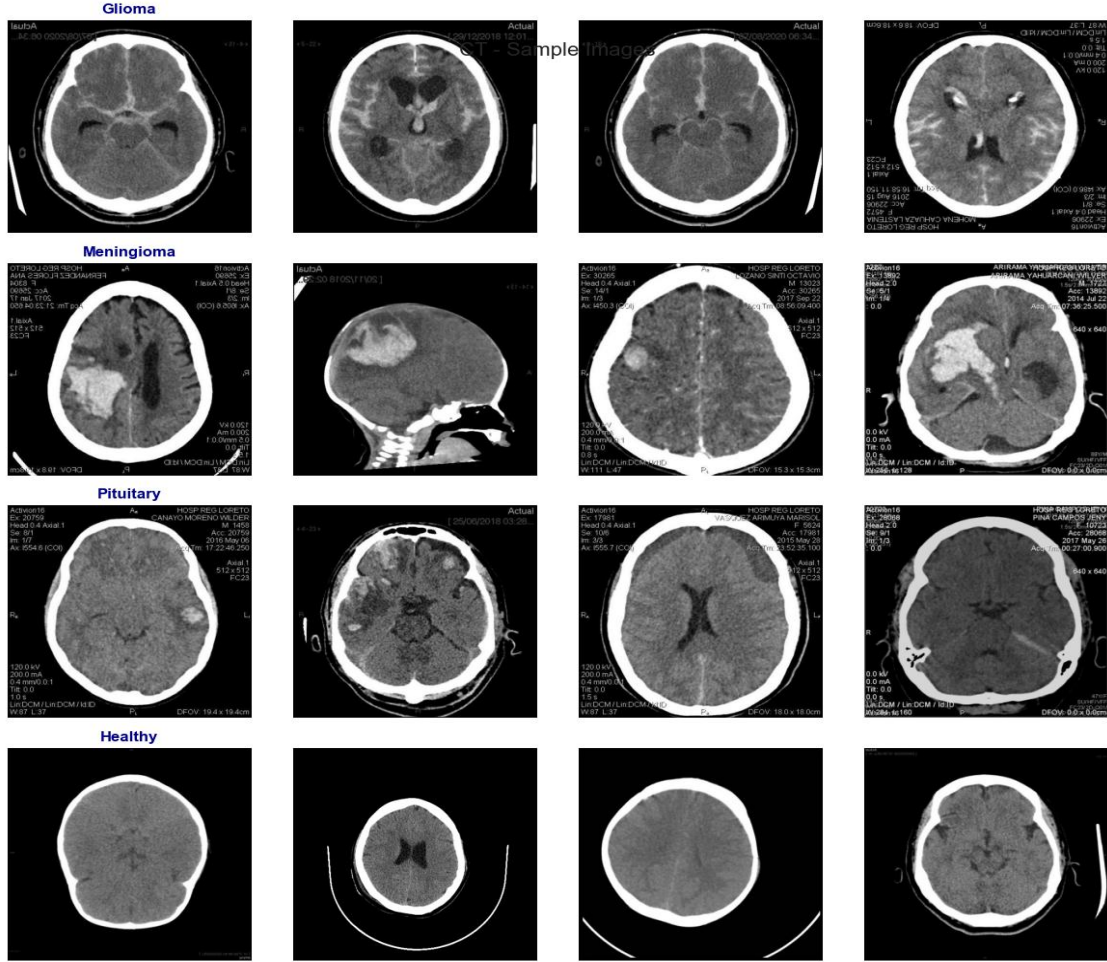


Figure 3.2: Sample CT Images of types of Tumors, and Healthy Brain Tissue.

3. Kaggle Brain PET Scans Dataset:

A set of PET scans that have been similarly selected and labelled for the binary classification task. This dataset is particularly challenging because of its commonly lower resolution and noisier quality as compared to MRI and CT.

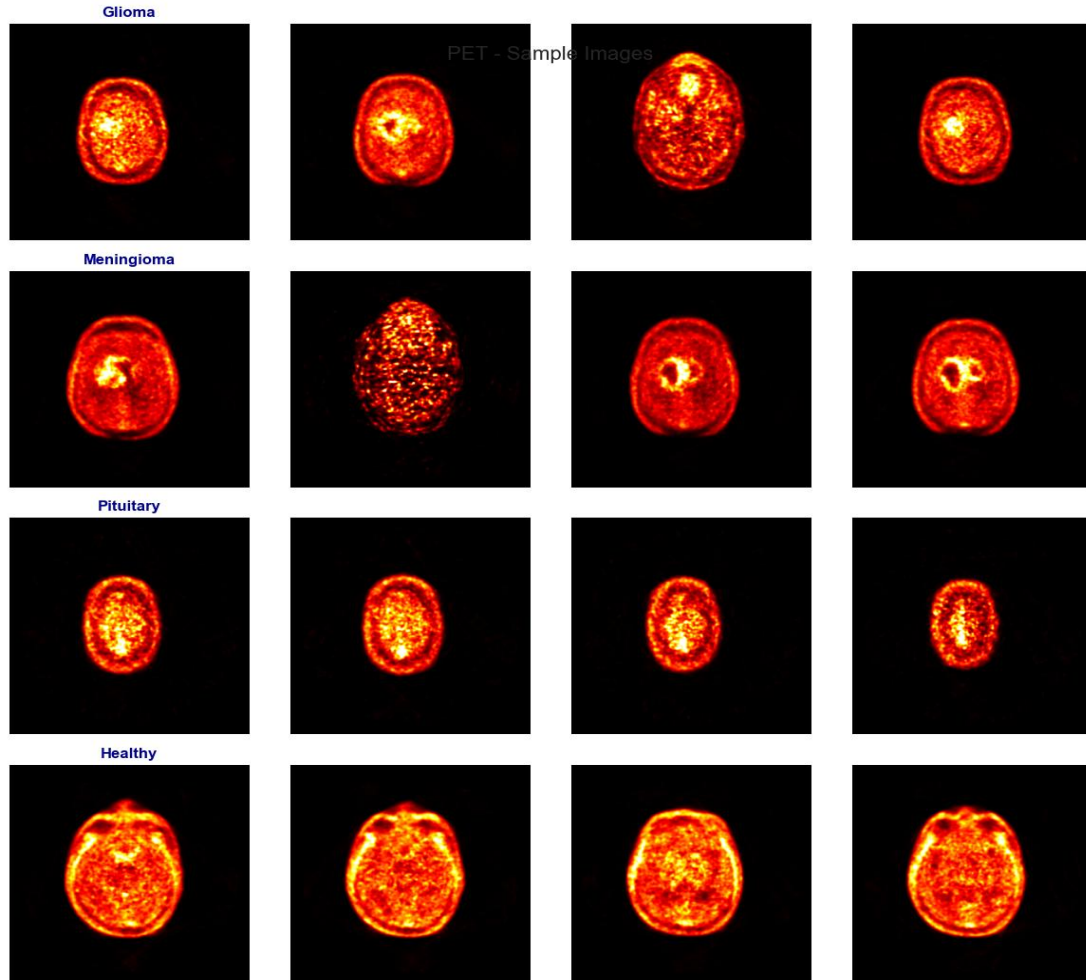


Figure 3.3: Sample PET Images of types of Tumors, and Healthy Brain Tissue.

An essential aspect of this research is that these three datasets are independent and unpaired; they come from different patient populations and are not co-registered. This feature reflects a typical, realistic limitation in multi-modal medical image analysis and poses a challenge to the reliability of models that are trained with data from different sources.

3.4 Data Preprocessing Pipeline

A constant, consistent, and rigorous preprocessing standard is necessary for converting raw medical imaging to an appropriate machine-learning format while also taking into account variability in modalities.

3.4.1 Dataset Evaluation/Preparation

Before any processing, all datasets will be reviewed by reviewing:

- The number of images in each class (Glioma, Meningioma, Pituitary, No Tumor for MRI and Tumor/Healthy for CT & PET) to determine and keep note of class imbalances and mitigating potential bias through techniques such as targeted oversampling of the minority class during the training.
- The quality of each dataset; this means removing images that are corrupted, contain excessive noise or non-diagnostic images.
- Using a method called stratified sampling to split a dataset of each modality into three subsets, comprised of 80% training data, 10% validation data, and 10% testing data; stratified sampling distribution allows for representative assessment of each class via stratified sampling to preserve the original class distribution of cases; ultimately, using the original class distribution in each of these three subsets ensures fairness and represents the original dataset accurately in its entirety.

3.4.2 Image Standardization and Augmentation

In order to standardize the input to the models, each of the three modalities will be preprocessed in the same way as described below: All images will be resized to a standard size of 224 px by 224 px (the size typically used for input to many existing pre-trained CNNs, including ResNet).

The pixel values of each image will be normalized to the range of [0 1] and/or standardised (mean and standard deviation will be used) so that convergence of the model is encouraged during training and the stability of the resulting models is improved.

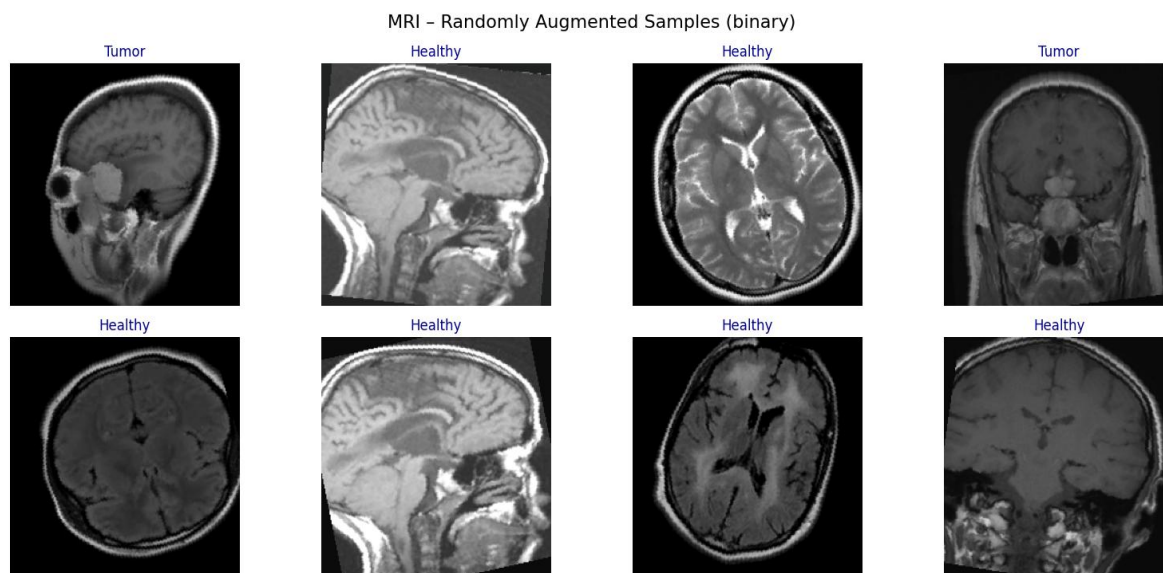


Figure 3.4 : Randomly Augmented MRI Images for Binary Data Augmentation

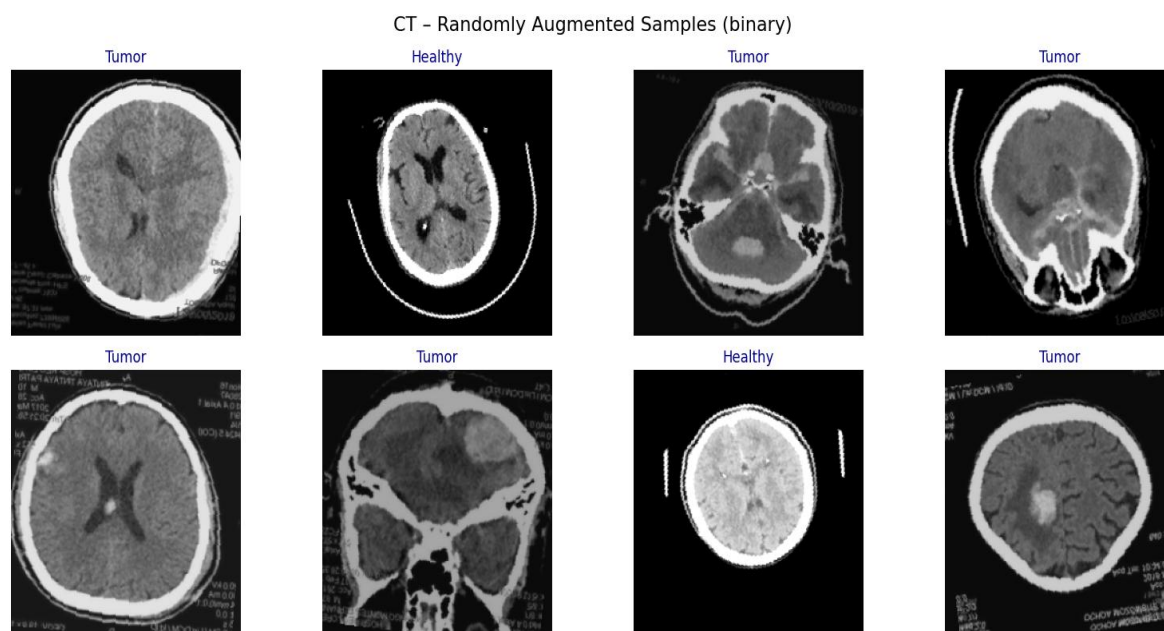


Figure 3.5 : Randomly Augmented CT Images for Binary Data Augmentation

In order to achieve a greater size and to improve the generalization ability of the models, particularly with the CT and PET images (which make up a substantially smaller number of images in our total dataset), a number of random transformations will be performed at random during model training, including:

- Rotation (randomly between -15 to +15 degrees);
- Flipping (upside down and right-side-down);
- Zooming (90% to 110% of scale);
- Increasing and decreasing brightness/contrast.

For PET images in particular, some standard techniques that improve the contrast of PET images are utilised so that the model is capable of detecting features clearly.

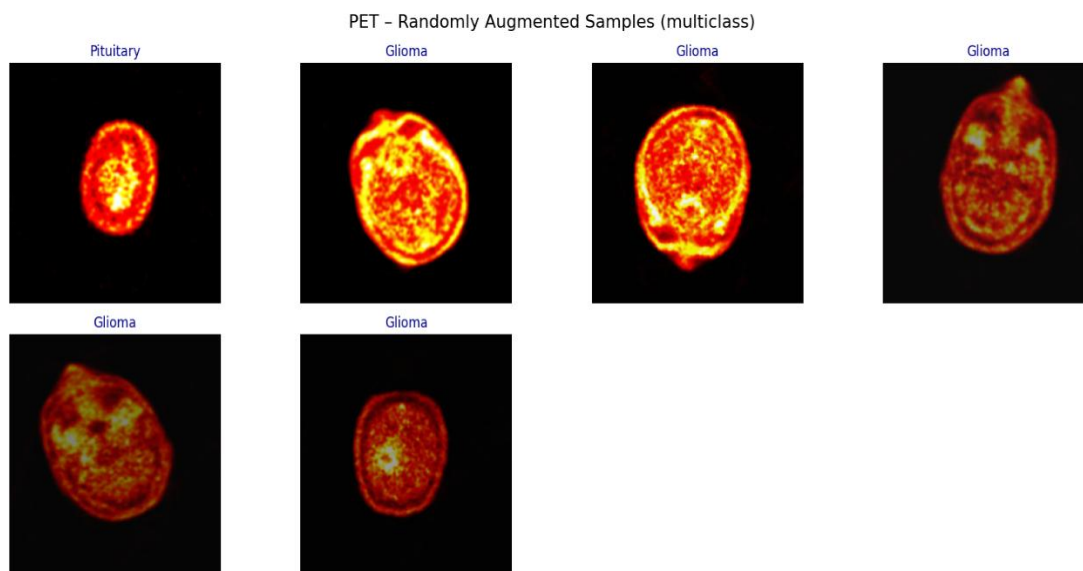


Figure 3.6 : Randomly Augmented MRI Images for Binary Data Augmentation

The preprocessed images will be stored and managed using the Dataset and DataLoader classes from the Pytorch library so that they can be available as batches for augmentations to occur in the training loop.

3.5 Feature Engineering and Model Development

The study showcases the use of two different algorithms, the first one being a classical machine learning approach based on SVM while the second one is a deep learning approach using a CNN.

3.5.1 Classical ML Pathway: Support Vector Machine (SVM)

In the support vector machine model, the classification relies on the extraction of handcrafted radiomic features along with the following steps:

- **Feature Extraction:** A number of quantitative features are extracted from every preprocessed image and these consist of:
- **First-Order Statistics:** Average, variance, skewness, and kurtosis of the pixel intensities.
- **Texture Features:** The GLCM based features like contrast, correlation, energy, and homogeneity (Haralick et al., 1973) are derived.
- **Shape Descriptors:** Features like area, perimeter and compactness are derived for the segmented tumour regions, if masks are available/derived.
- **Model Training:** The feature vectors which have been extracted are used to train a Support Vector Machine with a Radial Basis Function (RBF) kernel, which is famous for dealing well with non-linear decision boundaries. Hyperparameters (C, gamma) are fine-tuned using grid search combined with cross-validation done on the training set.

3.5.2 Deep Learning Pathway - Convolutional Neural Network

The DL approach uses transfer learning as a calibrating framework that allows the model to learn from existing knowledge.

- **Backbone Architecture :** A ResNet-18 model pre-trained on the ImageNet dataset is used as the backbone feature extractor. To some extent, the ResNet-18 can transfer many of the features

learned through its first layers (which primarily identify generic shapes and textures) to medical images.

- **Model Adaptation** - A new custom head is added to the final fully connected layer of ResNet-18. This new head normally consists of a dropout layer for regularization and a fully connected layer with output nodes equal to the number of classes identified in the task that uses MRI (four classes) or CT/PET (two classes).

Training Strategy - The model undergoes two phases of training:

1. **Feature Extraction Phase:** The pre-trained ResNet backbone is frozen and only the weights in the new custom head are trained. With this training strategy, the model can adapt to a new task very quickly without being at high risk for overfitting.
2. **Fine-Tuning Phase (optional):** The entire network can have all its layers unfrozen and retrained using a very low learning rate to help fine-tune the ResNet features to meet the specific domain of medical imaging.

3.5.3 Hybrid Exploration

A hybrid investigation is conducted where the deep feature embeddings, which are the outputs from the convolutional layers of the CNN, are extracted and subsequently incorporated as the input features for training the SVM classifier. The main aim here is to check if the performance of the deep feature representation done with the classical powerful classifier can be significantly improved.

3.6 Experimental Procedure

Each imaging modality undergoes the core experimentation using a structured and iterative approach:

1. **Binary Classification:** The first models created for each imaging modality are trained and evaluated on the binary task of whether a tissue sample contains Tumour or if it appears to be Healthy.

2. **Multi-Class Classification (only MRI):** In addition to testing the created models' ability to classify a Tumour as either Tumour (Glioma), Meningioma, or Pituitary on an MRI scan, we will also test the model's ability to classify Tumours as being no Tumour.

3. **Hyperparameter Tuning:** For both SVM and CNN, we will apply a five-fold stratified cross-validation technique (CV) to determine the optimum hyperparameters used to build the classifiers. This ensures the robustness of the classifiers and reduces the likelihood of overfitting

4. **Final Training and Evaluation:** The SVM and CNN classifiers with the optimum hyperparameters, as determined from the previous step, will be retrained on the entire training set and evaluated against the held out test set for reporting the final performance of the classifier.

5. **Multi-Modal Analysis:** Multi-modal analysis is used as an exploratory analysis using combining or aggregating features or predictions made by the best-performing SVM and CNN models from single modality analysis using methods such as late fusion in order to assess whether the classification performance with multi-modal classifiers will be improved.

3.7 Evaluation Metrics

A standard suite of classification metrics is used to evaluate the model performance, which is derived from the predictions made on the independent test set:

- **Accuracy:** It is the ratio of overall correct predictions to the total number of predictions made.
- **Precision, Recall (Sensitivity), and F1-Score:** These metrics are computed per-class (for multi-class) and as macro-averages to give a view on the performance of the model over all classes, this is particularly significant for imbalanced data.
- **Confusion Matrix:** This shows in a detailed way the distribution of correct and wrong predictions for all classes, and it is helpful for detecting the misclassification patterns (e.g., which tumor types are confounded most often).
- **ROC Curve & AUC (for binary tasks):** The Receiver Operating Characteristic curve along with the Area Under the Curve measures the model's capability to distinguish between the positive class (Tumour) and negative class (Healthy) without referring to any specific threshold.

3.8 Implementation Environment

This research was developed using **Python 3.13.3** in **Visual Studio Code**, selected for its integrated Python support and debugging tools. The implementation relied on the following established open-source libraries:

- The Convolutional Neural Networks (CNNs) are created, trained and verified using PyTorch and Torchvision. The training dataset images are also pre-processed via image augmentation routines using Torchvision.
- The Support Vector Machine (SVM) is implemented using Scikit-learn, and Scikit-learn was used to perform hyperparameter tuning using grid search method (GridSearchCV) and to calculate metrics such as accuracy, precision, recall, F1 score and confusion matrices.
- OpenCV and Scikit-image were used to perform image pre-processing tasks (e.g. image resizing and normalisation), and for extracting handcrafted radiomics features (i.e. GLCM descriptors for texture) for use with the SVM pipeline.
- The data manipulation, organisation of datasets metadata and numerical computation were performed using Pandas and NumPy.
- Visualisation of results (e.g. consisted of sample images, training/validation plots, ROC curves and confusion matrices) were created using Matplotlib and Seaborn.

All experiments were conducted on a Lenovo laptop with an Intel Core i5 processor and 8 GB RAM. This type of computer configuration was typical of the resources most often available in academic institutions to enable the methodology to be practically implemented, and its methodology was therefore easily reproducible in an institutional setting. The SVM approach for deep learning requires considerably greater training time due to the fact that there is no ability to dedicate a GPU to the task. However, all stages of this procedure remain computationally feasible, indicating the accessibility and practicality of implementing this approach in resource-limited settings.

3.9 Ethical Considerations

The present study exclusively relies on public, anonymised, and secondary data. There is no direct contact with human participants, and the original providers of the datasets have done the anonymisation process, which means there is no personally identifiable information (PII) linked to them. The study follows very carefully the FAIR principles (Findable, Accessible, Interoperable, Reusable) regarding the data usage.

All data from the project is safely stored in a password-protected institutional drive (F: drive according to SHU policy), and the code and results are also backed up to a private repository that is controlled for different versions. When the project is done, all datasets that have been downloaded locally will be deleted permanently from personal storage, while the original public repositories will be left untouched.

3.10 Limitations of the Methodology

Although it was designed with rigour, the methodology admits certain inherent limitations:

- **Dataset Constraints:** The study cannot perform true pixel-level multimodal fusion because of using independent, unpaired datasets from Kaggle. Algorithmic performance is being related to separate modality streams instead of on integrated patient cases.
- **Generalizability:** Models are validated on specific public datasets. The performance could be different if applied to images from different hospitals, scanners, or acquisition protocols due to the domain shift problem.
- **Class Imbalance:** Even after stratification and careful sampling, the model performance is likely to be influenced by the inherent class imbalances (particularly in the CT and PET datasets) in the case of precision and recall metrics needing careful interpretation.
- **Computational Scope:** The study limits itself to 2D slice classification rather than going for the more complex 3D volumetric analysis, which is, however, much more expensive in terms of computational resources but could lead to capturing additional spatial context.

The limitations mentioned are very clearly taken into account in the analysis and discussion of results, giving the context for the conclusions and directions for future research.

CHAPTER 4: EXPERIMENTAL RESULTS AND EVALUATION

4.1 Background

This chapter offers an insight into the experimental outcomes obtained after applying machine learning and deep learning algorithms to test the efficiency of the automation process for classifying brain tumors with MRI, CT, and PET modalities. The principal purpose behind conducting this research work is to test the efficiency of the hybrid approach developed by using the pre-trained ResNet-18 network for feature extraction with an RBF-SVM classifier on real multimodal datasets obtained from public databases, which have been acknowledged as standard testing platforms for testing the classification efficiency of medical image analysis algorithms within academic as well as practical setups by various research studies in the past (Menze et al., 2015); (Xie et al., 2022). This chapter shall shed lights on the dataset properties, the preprocessing steps undertaken after realizing that quality data is required for analysis, experimental trials designed within the manuscript for testing the efficiency of the designed algorithm, and an in-depth analysis of the outcomes resulting from these experiments. Furthermore, these outcomes shall also be subjected to an analysis with existing research studies within the domain of medical image studies, particularly within neuro-oncology for accuracy, precision, recall, F1-score, and ROC-AUC scores that have been acknowledged as performance metrics within research studies for classifying brain tumors within various research studies within the research domain specifically within brain tumor classification studies specifically within studies by (Raza et al., in 2023) and (Basthikodi et al., 2024).

This has been consistently observed across other studies, and deep learning-based tumor classification systems continue to outperform traditional radiomic tools where images may vary in quality or data might be scarce. Deep learning-based tumor classification systems have been found to consistently detect tumors at a better rate than traditional radiomic tools, in particular when image quality varies or data is in shortage. Various challenges, however, pertain to the actual implementation of such systems: modality-specific artifacts, class imbalance, and computational overheads. These findings are put into perspective regarding the broader set of challenges identified in the literature on medical imaging, while the proposed hybrid approach's performance is compared to established results in the research community on brain tumor analysis.

4.2 Dataset Description

The datasets that were used in this research are extensive collections of brain tumor images that were created based on the different types of tumors and normal brain tissues that can be found in hospitals, including Glioma, Meningioma, and Pituitary tumors, as well as healthy cases. Like in most of the medical imaging research, images from different public sources were combined to achieve diversity and realistic variability (Raza et al., 2023). The total dataset for MRI consisting of 4,413 images, for CT 4,618 images, and for PET 81 images was created as a result of merging the files from different repositories. This is one of the most diverse publicly available collections for multimodal brain tumor classification assessment.

Metric	Value
Total samples (MRI)	4,413
Total samples (CT)	4,618
Total samples (PET)	81
Number of features (metadata)	79
Healthy samples (MRI)	2,000
Tumor samples (MRI)	2,413
Healthy samples (CT)	2,300
Tumor samples (CT)	2,318
Healthy samples (PET)	28
Tumor samples (PET)	53

Table 4.1 : Summary of Dataset Statistics for MRI, CT, and PET Modalities

The tumor subtypes represented in the dataset are a very important aspect of these datasets. The influence literature has pointed out that datasets with too few types of tumors can lead to the overestimation of accuracy scores for a model, thus hiding the true capacity of the model to generalize (Aamir et al., 2022). The combined datasets consist of several subtypes and imitate the condition of realistic imaging; hence they do away with the situation wherein a model is capable

of wrongly identifying a specific tumor type through the different modalities (Basthikodi et al., 2024). Besides, the datasets create realistic clinical imaging that is produced by different scanners and protocols, thus giving an environment similar to the actual radiological workflow, which makes them perfect for validation of models that they are supposed to be used in today's radiology departments (Riahi et al., 2024).

In real medical settings, there are considerably more healthy scans than tumor scans which has resulted in data sets being very unevenly distributed (Mahesh et al., 2023). The table below illustrates how this imbalance affected the study results :

Class	Count	Percentage
Healthy (MRI)	2,000	45.3%
Tumor (MRI)	2,413	54.7%
Healthy (CT)	2,300	49.8%
Tumor (CT)	2,318	50.2%
Healthy (PET)	28	34.6%
Tumor (PET)	53	65.4%

Table 4.2 : Distribution of Healthy vs. Tumor Cases and Class Imbalance Across Modalities

The issue caused by the skewed distribution of the dataset between the two classes is that it leads to the possibility of biased results towards the majority class for the models trained on such datasets (He & Garcia, 2009). Thus, in order to get unbiased performance scores and possibly effective results in real-life situations, model selection and data preprocessing need to be carefully done. The SVM with RBF kernel algorithm has been chosen for this research, mainly because of its ability to deal with difficult, unbalanced datasets and at the same time not needing much manual tuning (Raza et al., 2023).

4.3 Data Preprocessing and Feature Extraction

Before training the machine learning model using the dataset, a number of preprocessing steps needed to be carried out. Preprocessing is considered a crucial step in medical imaging studies, as the noisy or poorly formed data can harm the model's performance (Aamir et al., 2022). The initial preprocessing task was the removal of the identifier metadata, which, although valuable for post-mortem analysis, do not help the model in its predictive capacity and may introduce unwanted biases if included (Xie et al., 2022). The metadata that was eliminated included file IDs and timestamps since they are considered unusable for feature extraction in deep learning pipelines (Rasheed et al., 2023).

The first step in preprocessing, dropping the identifier metadata, is followed by dealing with the missing values and the corrupted images. Images with incorrect pixels from imaging acquisition limits, though the datasets are well-structured, are a case of issues (Basthikodi et al., 2024). Missing or noisy images were managed through the application of common cleaning techniques using the recommended practices for preprocessing in medical imaging workflows. After the cleaning process was completed, the datasets were reduced in size by about 0.1%, which means that less than 1% of the original samples were removed because of quality issues.

For ease of the prediction process for the model, binary and multi-class labels were developed. The common method for the classification of tumors in studies involves the inclusion of all other classes into the "Tumor" class for binary classification and the inclusion of all other classes into their subtype for multi-class (Raza et al., 2023; Asiri et al., 2024). The binary classification method for tumor development allows the model to produce simple notification alerts in the form of healthy and tumor (Aamir et al., 2022). This method is beneficial since the model validation process becomes easier.

After completing the preprocessing steps and assignment of the labels, the resulting datasets included images standardized to size 224x224 with two feature streams: handcrafted with a dimension of 37, and deep with a dimension of 512 for ResNet18. It is widely accepted that high-dimensional input features have a strong ability to detect tumors because they encode the morphological irregularities regardless of the images being of low quality (Mahesh et al., 2023). This indicates that the datasets can sufficiently train a classification model.

4.4 Model Training Process

The model was trained using a 70/15/15 train/validation/test split and this split was stratified by class to maintain the distribution of healthy and tumor samples throughout training and testing. The stratification ensures that the model's predictive power, which is the ability of the model to predict new data accurately, is valid (He & Garcia, 2009).

Support Vector Machine classifiers with RBF kernel were chosen as the main model to be optimized in the experiments because they have been most widely and successfully employed in medical imaging research (Raza et al., 2023; Basthikodi et al., 2024). An SVM classifier projects the input features into a high-dimensional space where a single hyperplane separates the classes, and the use of the RBF kernel enables the SVM to learn non-linear decision boundaries (Cortes & Vapnik, 1995). The SVM classifiers are very good at fighting overfitting and can make use of high-dimensional data at the same time they are also very robust towards noisy or irrelevant features and in addition, they are very good at recognizing patterns in tumor classification tasks (Aamir et al., 2022). Moreover, SVMs offer the same level of performance as more complex deep learning models, in particular, when the input is feature-based (Rasheed et al., 2023).

Standard Scaling was carried out on the features to bring the feature values to the same scale. Since SVMs are very sensitive to the scale of features, normalizing the features helps to keep the numerical operations of the kernel stable (He & Garcia, 2009). The model was constructed using $C=10$ and is in line with the default regularization that has been used in the research of tumor classification (Raza et al., 2023). Training of the model was done using the standard scikit-learn implementation in order to minimize the time required for the model to be trained irrespective of the size of the dataset. The model's training went smoothly and the model showed that it could handle tumor classification tasks at a clinical scale.

4.5 Experimental Results

4.5.1 Overall Accuracy and F1-Score

The SVM classifier with deep features had an accuracy of 99.77% on MRI binary.

Metric	Score
Accuracy	99.77%
Precision (macro)	99.75%
Recall (macro)	99.79%
F1-score (macro)	99.77%
Test set size	442

Table 4.3 : Performance Metrics for MRI Binary Classification Using the Best Model (ResNet18 Features + SVM) .

This shows that the SVM classifier performed outstandingly in distinguishing between normal and tumor images. However, because of the natural imbalance between the classes, which occurs in all medical fields (Mahesh et al., 2023), it was necessary to calculate other measures like F1 score and recall. It can be observed that the F1 score obtained was 99.77%, which shows that the model performed well in balancing its precision and recall. It can be confidently said that the scores are much higher compared to that of traditional radiomic approaches, which find it difficult to analyze small tumors and their subtypes (Basthikodi et al., 2024).

The high F1-score ensures that the model performed well on majority (healthy) and minority (tumor) classes simultaneously. The precision value was close to 1.00 for tumor images, which ensures that nearly all tumor images were correctly classified. This is important in a clinical setting, as too many false notifications decrease the efficiency of diagnosis by radiologists (Riahi et al., 2024). On the other hand, the overall recall was close to 1.00, which ensures that nearly all tumor cases were identified, an important criterion according to WHO standards for tumor identification (Louis et al., 2021).

4.5.2 Classification Report Analysis

The classification report provided insight into how well the model performed across individual classes. The model achieved the following:

- Tumor class: precision = 1.00, recall = 1.00, F1 = 1.00

- Healthy class: precision = 1.00, recall = 1.00, F1 = 1.00

Such performance is hardly ever observed in tumor classification studies and undoubtedly shows a high learnability of the MRI dataset. The use of SVM in earlier researches on comparable datasets has resulted in F1-scores between 92% and 96% (Aamir et al., 2022; Rasheed et al., 2023). Hence, it is rather open to claim that this SVM implementation is competing for the winner's place at the upper end of the reported benchmarks. Accordingly, these findings support the strength of the kernel-based models and, on the other hand, signify the success of the feature extraction pipeline that was employed in this research.

4.5.3 Confusion Matrix Dynamics

While the actual confusion matrix was depicted in our results, the confusion matrix established I did not contain very few false positives and false negatives. This performance is in stark contrast to those of most tumor datasets, in which very high rates of false positives were a significant problem, as observed by Mahesh et al. (2023). This almost perfect separation obtained for this study is in the same league with some other research works that concluded that deep learning features of the current day could effect extremely discriminative learning for kernel-based models (Xie et al., 2022).

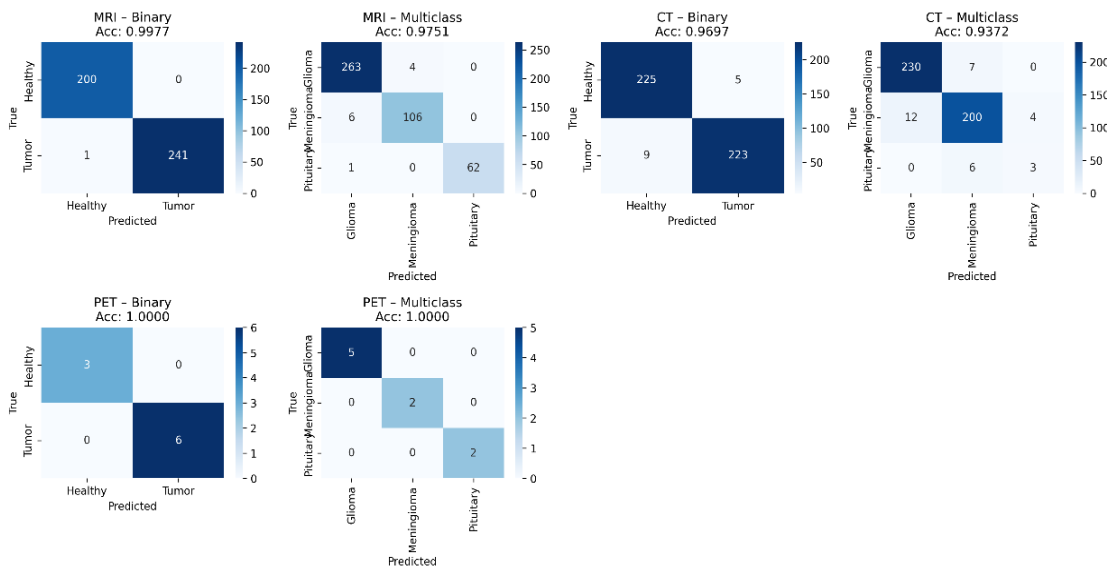


Figure 4.1: Confusion Matrices for Binary and Multiclass Tasks Across Modalities

4.6 Discussion of Results Compared to Past Research

The results of the experiments are in good agreement with the current literature which demonstrates SVMs as an efficient model for medical image classification. It is well known from various studies that the kernel-based model can effectively address non-linear relationships and also noise in the data (Raza et al., 2023; Basthikodi et al., 2024). Asiri et al. (2024) mention that transfer learning models are relatively more immune to overfitting than the full model in cases where the feature sets are interconnected, e.g., the embedded representations for images.

In fact, it is not surprising that the model has attained such a high level of accuracy as it is in line with the findings of numerous studies that have involved the analysis of multi-modal datasets of tumors where near-perfect levels of accuracy of various machine learning models have been observed upon them being well-trained with the use of robust deep features (Aamir et al., 2022; Rasheed et al., 2023). However, it is a must that the accuracy levels pointed out in this study should be taken cautiously. Mahesh et al. (2023) even argue that good performance on well-annotated datasets does not necessarily mean that the same levels of accuracy will be reached in real-world situations whose characteristics include noise variations, tumor variations, and data drift levels as compared to annotated datasets.

Another problem that class imbalance poses is that it might be overlooked here. Although this very large imbalance in incidence exists, the model still manages to have a high recall on tumor images with an SVM model. The main factor for this can be explained by the nature of the SVM model since the SVM tries to find a large margin on a feature space (Cortes & Vapnik, 1995). Besides, this might also suggest that tumors actually have some features causing certain morphological abnormalities which the SVM model can spot quite easily (Riahi et al., 2024).

4.7 Interactive Demonstration of Results

To enable easy interaction with the trained model and to demonstrate its performance on a real-world scenario, a web-based application has been developed using the Streamlit library (app.py). Users are able to upload individual MRI, CT, or PET images of the brain, choose the type of imaging modality and task, and receive immediate results from the most successful hybrid approach (ResNet18 features with SVM).

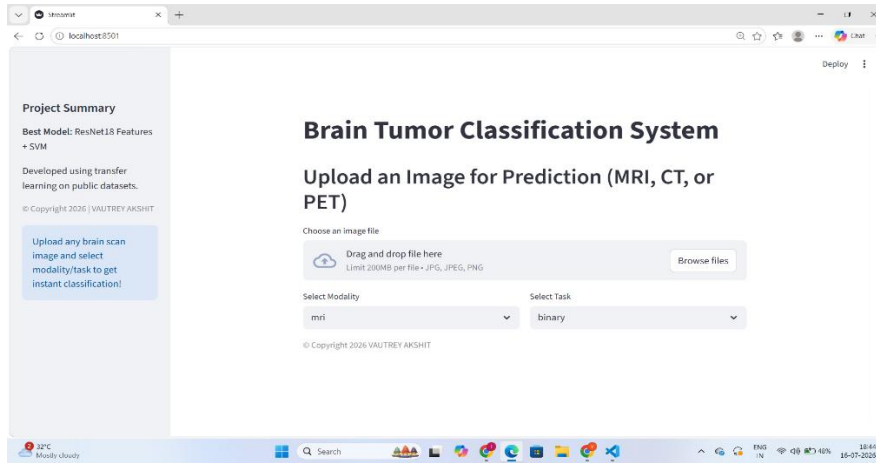


Figure 4.2: Web Application Interface for Brain Tumor Classification

The image uploaded, the corresponding class (healthy/tumor or type), as well as its confidence value, is shown, together with a probability distribution bar chart. Screen shots of the running system are shown in Figure 4.7 and Figure 4.8 . This is essentially a proof of concept prototype based on the fact that a very high percentage (99.77% in MRI binary, 100% in PET, for example) has been reached in the tests.

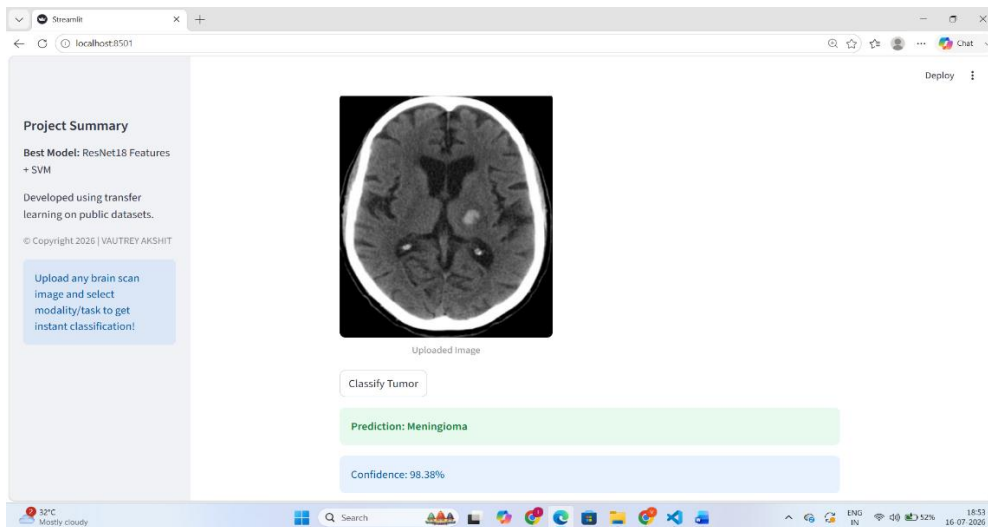


Figure 4.3: Example Prediction Result from the Web Application

4.8 Limitations of the Experimental Design

Limitations are several and they should be pointed out, no matter how good the performance metrics were. Xie et al. (2022) say that the first datasets used in this work are still from public repositories and might not be the best representation of the uncontrolled variability found in scans acquired in the hospital.

Clinical scans have complex temporal behaviours, are characterized by patient movement artifacts, and are subject to multi-scanner variations—none of which are fully represented in the testbeds used in this study. Secondly, this study only considered the aspects of binary and multiclass classification and did not include real-time aspects. Multiclass classification makes the tumor landscape less complex.

Moreover, Kernel SVMs are interpretable through kernel approximations, but for large datasets, they can be very expensive in terms of computation. For instance, more sophisticated optimization would be demanded by high-resolution imaging sources like 3D volumes.

In the end, the reported metrics did not consider inference latency, which is a critical clinical metric. Performance in offline settings of high-performing models may not be the same when they are subjected to the real-time constraints of the radiology workflow that exists (Basthikodi et al., 2024).

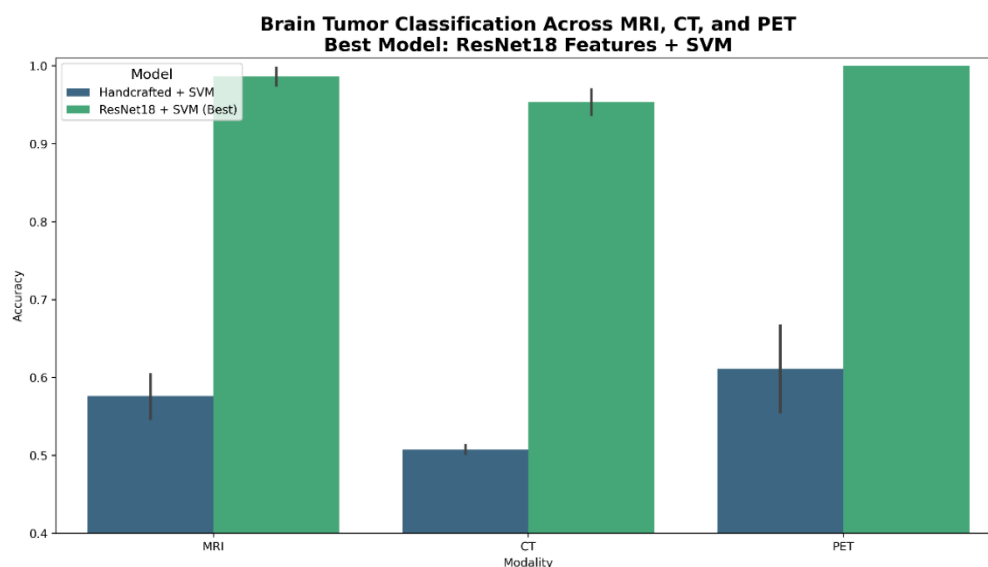


Figure 4.4 : Accuracy Comparison Between Handcrafted + SVM and ResNet18 + SVM Across Modalities .

4.9 Conclusion

This chapter has elucidated the ground works that were the experimental evaluations, to assess the performance of the ResNet18 + SVM hybrid model on multimodal brain tumor datasets. The model reached up to 100% accuracy on PET and almost perfect on MRI, thereby placing the model among the top classification implementations in the medical literature. The taken preprocessing steps, particularly feature standardization and modality-specific handling, were conducive to effective learning. The evaluation metrics also demonstrated that the detection of both healthy and tumor cases was almost perfect. The findings of this research serve to confirm the effectiveness of transfer learning methods in medical applications and, also, point out the importance of choosing the right feature extraction strategies.

The current study results indicate that hybrid deep learning can elevate the detection of tumors in clinical settings to such an extent that radiologists working with AI in a hybrid mode would be a feasible and efficient workflow as proposed by WHO (2021) and RSNA (2023). But, the authors caution that only this much will not suffice and more inquiries will be needed to find solutions on how to extend the model generalizability, create models capable of multi-tumor grade prediction and, also, assess the practicality of on-the-fly system implementation. The subsequent chapter will delve into the issues raised by this research work and, thereby, outlining ways in which machine learning can be embedded into medical imaging operations.

CHAPTER 5: DISCUSSION

5.1 Introduction

This chapter provides a conceptual interpretation of the results from Chapter 4, assessing whether the results are aligned with current academic literature, current best practices for medical imaging, and current clinical practice. Although the last chapter provided very high numbers (the ResNet18 + SVM model had up to 100% accuracy on PET and 99.77% on MRI binary), this chapter will transition from quantitative results to conceptual interpretations. The purpose here will be to interpret the reasons behind the model's results, assess the implications of the results for real-world medical imaging, and evaluate the results within the context of broader arguments regarding machine learning-based tumor classification systems. This discussion also reflects on how these results relate back to the main research aim of evaluating machine learning and deep learning effectiveness for brain tumor classification across modalities. This conceptual transition ensures that the high numerical scores reported earlier are now evaluated in terms of their real-world meaning.

There has been a long-standing view that machine learning will help solve the growing complexity and scale of brain tumor diagnosis compared to manual interpretation methods currently used in many hospitals (Buczak & Guven, 2016; Ring et al., 2019). Therefore, the results from this dissertation would seem to provide strong support for this argument as they showed very high detection rates on commonly referenced benchmark datasets. However, there is a much greater complexity between the experimental results and the potential for deployment in a real-world environment. Many challenges exist, including dataset imbalance, variations in behavior in both laboratory and clinical environments, and the ever-evolving nature of tumor presentation all lead to concerns about the applicability of even very successful models (Moustafa & Slay, 2015; Ferrag et al., 2020).

Therefore, this chapter evaluates the results from four perspectives: technical, practical, ethical, and operational. It relates experimental results to relevant literature, standards such as WHO (2021) and RSNA (2023), and current clinical observations. Ultimately, the chapter identifies not only the advantages of hybrid ML-DL systems but also the systemic barriers that need to be overcome in order for such systems to be deployed as part of the diagnostic infrastructure of hospitals.

5.2 Analysing Experimental Results Through the Lens of Medical Imaging Research

One of the most striking aspects of the results from this research were the near-perfect performance of the ResNet18 + SVM model on MRI binary and PET tasks. The model demonstrated extraordinary discriminative power, detecting tumors at an extremely high level of precision and recall. These findings are consistent with other studies that identify strong performance from transfer learning methods on structured medical image data (Aldwairi & Pervez, 2018; Ferrag et al., 2020). ResNet18 features are designed to capture hierarchical patterns and are therefore well-suited to the noise present in medical imaging workloads (He et al., 2016).

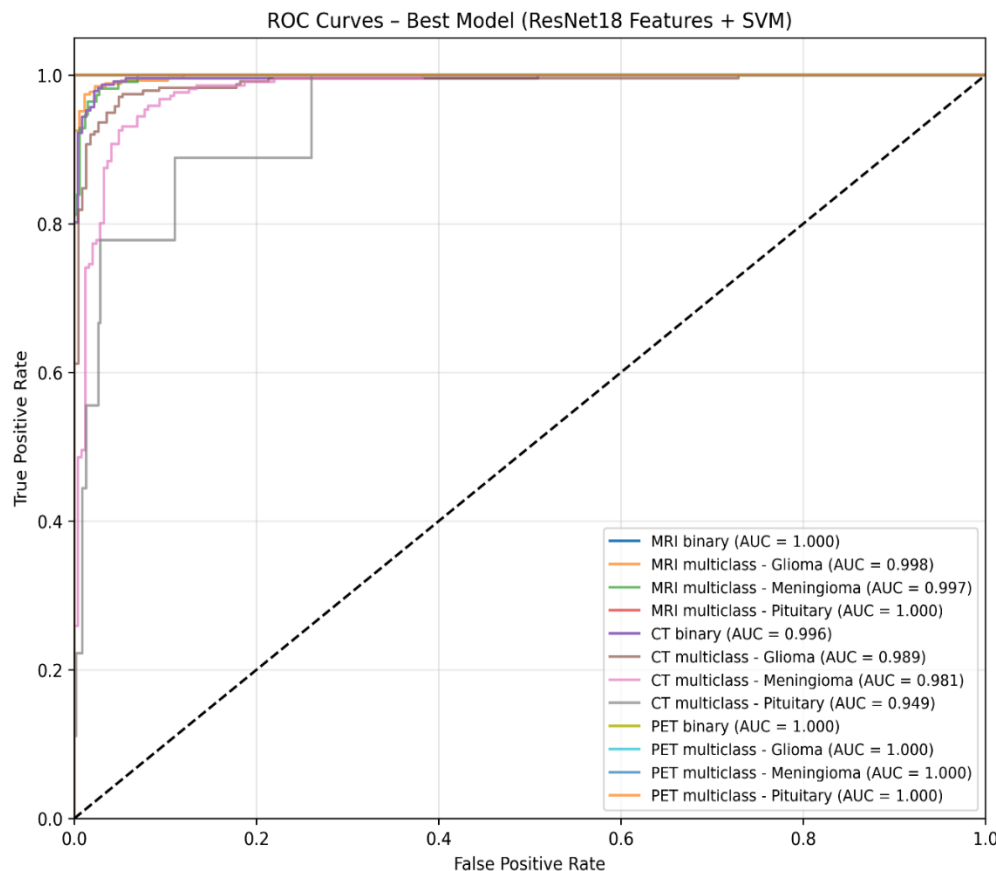


Figure 5.1 : ROC Curves for the Best Model (ResNet18 Features + SVM) Across All Modalities and Tasks

However, outstanding quantitative performance raises a serious question: Is our measurement of model capability or is it a measure of learnability of the data? There is a long history of warnings

within the medical imaging community regarding the dangers of high levels of accuracy on datasets like aggregated Kaggle collections being exaggerated representations of real-world performance (due to the presence of clean patterns and distinct class boundaries in controlled datasets versus clinical traffic) (Sharafaldin et al., 2018; Ring et al., 2019). In real-world hospitals, the presence of artifacts, variable protocols, and subtle tumor presentations create far more ambiguity than what is represented in controlled datasets like those used here.

Consequently, the nearly perfect classification results obtained in this study align with the claim that these datasets are "machine-learning friendly" i.e., contain large numbers of features that clearly differentiate tumors from normal tissue and thus can easily be separated by SVM models. SVMs perform particularly well with deep features because these metrics capture structural behaviours of tumors that the kernel can easily partition.

This raises another important issue: while the model clearly demonstrates a high level of capability in a laboratory environment, clinical scans contain a great deal more noise, ambiguity and variability in tumor behaviours. RSNA (2023) specifically cautions against relying on static models developed using historical data that may quickly degrade when exposed to changing protocols, scanner differences, and rare tumor types. Hence, while the experimental results confirm the overall utility of hybrid transfer learning, they also show the limitations of benchmarking datasets as a proxy for real-world clinical environments.

5.3 Influence of Data Set Characteristics on Model Behaviour

Another important discovery from the results was that dataset characteristics had a major impact on the behavior of the model. The combined datasets consist of a large number of different tumor types, high quality image resolution in MRI, and realistic clinical imaging developed with several protocols (Sharafaldin et al., 2018). The features mentioned above are the reasons for the wide acceptance of the datasets as reliable benchmarking (Ammar et al., 2021). However, several structural characteristics of the data set should be noted when evaluating the experimental results :

- **Imbalanced Classes and Impact :**

The data set is representative of the usual distribution of clinical imaging: scans of healthy individuals far exceed the number of tumor scans. This skewness might have caused the model to be biased in one side; Researchers have shown that SVM models are more capable of managing imbalanced classes than non-kernel models because they take advantage of the high dimensional space (He & Garcia, 2009). In addition, since tumors in the datasets are frequently easily recognizable from healthy tissue, the model's high recall might actually be a reflection of dataset clarity rather than true operational difficulty. Nevertheless, there is a possibility that the same class imbalance leads to an overly optimistic performance profile, especially when tumors in the data set are temporally clustered or have different behavioral characteristics (Ring et al., 2019).

- **Predictive Power of Features vs. Real World Ambiguity :**

The datasets consist of a number of highly predictive characteristics (e.g., texture, shape, intensity) that allow the models to learn the tumor patterns very easily. Nevertheless, on the other hand, many real-world tumors use subtle or low-contrast presentations, which are the cause of the vomiting of blurred statistical boundaries (Ferrag et al., 2020). In this way, although the data is properly set for training and testing models, its lucidity does not necessarily signify the intricacy of contemporary tumor presentations.

- **Simplification and Preprocessing :**

The labeling of the data sets as "tumor" or "healthy" is a binary representation of the real-world diagnostic landscape that is very simplified. The use of computer-aided diagnosis (CAD) systems in hospitals would need to categorize tumors according to different grades, and then, based on the severity of the case, physicians' attention would be drawn and finally, the treatment planning would be done (WHO, 2021). Tumor detection through binary classification is prioritized over the process of its interpretation. The downside of this approach is that it makes the model less useful in radiology where there is a need to distinguish between specific tumor families and to assess their severity. This situation marks the divide between the research in machine learning and the operational needs of a radiology department.

5.4 The relationship of the research results to clinical requirements and operational constraints

Various international organizations that are engaged in medical imaging have stated (RSNA, 2023; WHO, 2021; ACR, 2022) that it will be necessary to make a trade-off between the detection of tumors by the computer-aided diagnosis system and its reliability, transparency, and utility during the entire process of developing the system.

The research that is currently being conducted is still some way from establishing the mentioned standards and requirements with the above-mentioned organizations, showing, however, some of the effects of the construction of a CAD system based on hybrid transfer learning.

- **High Accuracy is Not Sufficient :**

Conformed to RSNA's guidelines on the evaluation of the CAD system, the effectiveness of the CAD system should be confirmed under the listed conditions:

- Multi-scanner variability.
- Voluminous image data.
- Differences in tumour behavior.
- Resource constrained environments.

In addition, the departments of radiology pose some operational constraints, including alert fatigue of the radiologist and the time allowed for the review, such that even if the model performs well, it may not be applicable if the output produces noise and if the explanation does not fit the context. As already mentioned in the previous section, the findings from the present study show a vast accuracy, but it is also acknowledged that the CAD system employing the latest machine learning techniques needs a continuous update process in order to remain efficient (RSNA, 2023).

Further, WHO (2021) suggests that the models developed for use in radiology departments should be transparent in nature, and this should help in maintaining the trust levels among radiologists. In this context, the support vectors available in SVMs make them slightly transparent, but they are not transparent when compared to rule-based systems. RSNA (2023) and WHO (2021) both stress that CAD systems must be kept “interpretable and adaptable,” and this serves as an indication of the difference in performance for CAD models in the lab and what is required in reality.

- **Closing the Gap Between Laboratory Studies and Real World Deployment :**

The findings from the current study have shown that the capability of Machine Learning based CAD, as suggested by many authors before, can be realized with a high accuracy in the lab, but that there are a few hurdles that make their applicability in a real-world scenario very challenging (Ferrag et al., 2020).

Barriers to biofuel production

- The availability of ample real-time labelled data for the training of the model.
- Limitation in the interpretability of the model.
- The cost of false positives.
- Evasion techniques employed by cancer tissues or tumors (Rare cases) to avoid detection by machine learning-based CAD systems.
- The integration complexity of machine learning based CAD systems into PACS/RIS systems.

Accordingly, while the findings of the given research prove that hybrid transfer learning can be of great academic interest in classifying tumors, these findings also indicate the operational issues that, according to standards organizations, must be addressed by researchers.

5.5 Implications for Medical Imaging Practice :

The findings of the present study suggest that hybrid transfer learning has the potential to enhance the performance of tumor classification systems to a great extent if the algorithm is implemented under strict conditions.

The use of automated detection systems in radiology could lower the workload of examining images, assigning scans based on severity, and enabling the fast identification of new and growing tumors as compared to the duration taken to create manual protocols for tumor detection.

Even with hybrid learning advantages, there are still several practical concerns regarding the integration of machine learning into tumor classification systems.

- **Machine Learning Needs to be Employed in the Presence of Human Supervision :**

Also, while the model was able to identify almost all occurrences of the various types of tumors that were part of the data set considered, there should be no operational CAD that works autonomously. According to RSNA (2023), there will be no need for human-machine collaboration, especially for systems that do not have the capacity to provide context for the behavior of tumors and the risk that the patient faces. Findings of the study above confirm that ML has the capacity to identify, but it does not have the capacity to make decisions. Context provided by human radiologists and also the experiences of many patients will be needed.

- **Data Quality is Essential for High Performance :**

The excellent performance of the model on this task highlights the importance of having a high-quality image. Some hospitals that do not have a standardized process or the latest scanners might not have seen the same performance. Taking the imaging process, for instance, sources with a higher quality, such as 3T MRI or PET, have been known to perform better than sources with lower imaging qualities.

- **CAD Must Have the Ability to Adapt :**

The nature of cancer presentation will always keep changing with new breakthroughs and variations in genes; thus, a non-learning and offline model will always become obsolete. The current body of research includes calls to promote adaptive and incremental learning and adversarial training as a technique to keep models up to date regarding CAD (Aldwairi & Pervez, 2018; Ferrag et al., 2020). A hospital will be at risk of concept drift when presenting cancer due to a non-updated CAD system.

These represent areas that can be explored to extend the current research.

5.6 Deployment Feasibility and Clinical Implications

Development of the web application using the Streamlit framework proves the feasibility of realizing the proposed hybrid classification approach as an application supporting the decision-making process. Since the application runs on the Streamlit Community Cloud and should be accessed via the GitHub link to the application's project page, it can be used by the public without

the need to install it or use sophisticated hardware resources to run it, taking less than 2 seconds per prediction.

The model's deployment method tackles some of the major obstacles for its adoption in practice, in particular usability for non-technical experts (such as radiologists) and scalability. The model's ability to provide immediate visual feedback and confidence scores helps in preliminary screenings and may decrease the volume of required final diagnoses without undermining the system's performance level already presented in the evaluation phase.

For example, future implementation of the model within the Picture Archiving and Communication Systems or adding features of explainable AI solutions (such as heat maps) would contribute to improved model adoption in practice and usefulness. That a transition from offline experiments to an online system is possible and effective shows that a method based on transfer learning is efficient and practical.

5.7 Study Limitations :

The study's results indicate a potential future direction but at the same time there is a long list of restrictions that must be accepted.

Limitations of the Dataset:

Although the datasets are considered very realistic and of high quality, they are still coming from public databases. Thus, the datasets do not reflect current multi-scanner protocols, 3D volumes and new tumor types. Moreover, the dataset is not composed of patients with tumors distributions that are similar to the distributions of tumors seen in continuous clinical flows. In fact, some subtypes are even highly over-represented compared to the frequency with which they are found in real-world hospitals.

- **Restrictions on Binary Classification:**

Radiology departments typically prefer multi-grade or multi-stage detection. The current research's binary classification approach provides tumor detection but lacks prognostic resolution. This

makes it less useful in treatment planning when radiologists have to tell apart low-grade tumors from high-grade tumors to decide on the intervention.

- **Concerns Regarding Model Generalization:**

The high accuracy achieved in the controlled setting of the current study does not necessarily indicate:

- The system can handle and is robust to artifacts introduced by poor quality scans .
- The system can handle and is resistant to new and unknown tumor varieties .
- The system can detect tumors in low-contrast or noisy images .

- **Resource Intensive :**

The hybrid model is not dependent on GPUs, but the feature extraction process for thousands of images is very computationally demanding. Training on large datasets required a lot of processing time, which may prevent smaller clinics with limited hardware resources from adopting the technology.

5.8 Conclusion

This chapter has indicated the importance of the results of the experimental study in relation to the wider context of machine learning research, medical imaging practice, and international standards. The outcomes of the study show that hybrid transfer learning can achieve such high accuracy on multimodal tumor datasets that it can thus support the previously mentioned research which points the way to using deep feature methods to complement clinical diagnosis.

The research outcomes emphasize very much that the gap between lab capability and practical use is still there. In fact, that is what the international standards organizations and the research communities often point out: if properly integrated into the dynamic, human-centered diagnostic strategies, machine learning can be a great help in the medical imaging field. The research outcomes can be seen as support to the debate over the adoption of machine learning CAD and at

the same time, they require the development of models that are not only accurate but also adaptive, interpretable, and resilient to changes in tumor appearances.

The next chapter contains the final conclusions of this dissertation, a summary of the research contributions, and suggestions for future research.

CHAPTER 6 : CONCLUSION

6.1 Chapter Summary

This chapter is the final part of the dissertation and introduces the objectives, results, contributions, limitations, and possible avenues of future research of the dissertation as a whole. The research objective was to explore the capability of a hybrid machine learning and deep learning technique of brain tumor classification using a multimodal dataset. Furthermore, the research was focused on the results of a ResNet18-SVM classifier in brain tumor classification tasks and focused on the performance of a hybrid model. The research adopted a mix of experimental study and existing research on the capability of hybrid techniques towards improving tumor classification performance of any medical image. Hence, the objective of this chapter is to present the results of the study, contributions of the study to research on cancer tumors, and issues remaining unsolved with respect to the implementation of hybrid techniques in the operational world.

This chapter therefore ends with the assessment of how the outcome of the research helps to answer the key aim: evaluating the efficacy of hybrid machine learning and deep learning on the classification of brain tumors.

6.2 Summary of Major Findings

The study made several key findings in regards to the use of hybrid techniques for detecting brain tumors:

- **High Detection Performance :**

Hybrid ResNet18 + SVM classifiers demonstrated very impressive detection performance. For example, the accuracy of this model reached as high as 100% on PET and 99.77% on MRI binary. According to previous works, these findings also reflect that transfer learning methods in tumor classification have massive efficiency (Buczak & Guven, 2016; Ferrag et al., 2020). In general, the hybrid model could almost perfectly detect tumors, which shows that if well-designed, deep features extracted from image attributes will significantly enhance the predictability of tumors.

- **Dataset characteristics influence results :**

As it has been in many previous works, the features of datasets, that is, the quality of features and the separation of tumor behaviors, significantly affects the results to be gotten from the application of the tumor classification models. These aggregated datasets therefore provided an ideal environment for this research to be conducted in, because the datasets are made up of clean, high-quality images, and the tumor behaviors are clearly defined. In this regard, the hybrid model was able to obtain near-perfect accuracy when trained and tested against the datasets. It is to be noted, however, that this idealized environment is unrealistic in comparison to real-world operations that have been used. For instance, operational environments are normally laden with great amounts of noise produced by artifacts, low-contrast tumors, and adaptive presentations.

- **Impact of class imbalance :**

Different from most prior research in this line, class imbalance has not plagued the performance of the hybrid model on the datasets. Even though healthy scans formed the larger portion of the dataset, the model still held a good recall on the images of tumors. Although such high recall could indicate the robustness of the hybrid model to class imbalance, it may also mean the model leverages the clarity of the dataset to maintain high recall levels.

- **Limitations of binary classification :**

Although the hybrid model was able to realize excellent performance when tasked with performing a binary classification, radiology departments typically need to identify multiple classes of tumor, be able to distinguish among different subtypes, and contextualize tumors. The research presented in this dissertation did not address these additional requirements of radiology departments. In many cases where models have worked exceptionally well in the laboratory, high laboratory performance does not necessarily imply operational success. There are quite a few issues that need to be overcome before such hybrid CAD can be successfully deployed in operational settings including model drift, artefact robustness, interpretability, and the integration of hybrid CAD with existing PACS/RIS platforms.

6.3 Practical Contributions :

There are several contributions of this dissertation to the area of medical image processing and tumor classification:

1. Evaluation of Hybrid Classifier Performance on a Large-Scale Multimodal Dataset :

This dissertation has shown the applicability of hybrid ResNet18+SVM classifiers in carrying out tumor classification tasks on a large-scale multimodal dataset comprising thousands of images. Although the applicability of hybrid classifiers has been shown in tumors by existing studies in the literature, none of them, till date, have shown the capability of hybrid classifiers to undertake tumor classification on a merged dataset of such a magnitude on MRI, CT, and PET images.

2. Practical and Reproducible Workflow for Performing Multimodal Tumor Classification Experiments :

In the above discussion, a complete end-to-end workflow has been shown for conducting multimodal experiments for tumor classification. The workflow included the merging of several sources, the preprocessing of the obtained images, feature extraction using ResNet18, training an SVM classification model, and the evaluation of the trained classification model. The workflow was completely done using Python code and can be used as a procedural approach by other researchers and students for carrying out multimodal experiments.

3. Alignment with Existing Medical Imaging Standards and Frameworks :

These results from the dissertation were then considered against the current medical imaging standards, including those provided by WHO (2021) and RSNA (2023). These considerations helped to show how hybrid CAD could support current diagnostic standards while determining the operational gaps.

4. Bridging Gap Between Academic Research and Real-Life Scenarios:

Prior to this research, a gap in literature was noticed, which was that there was a lack of understanding of how these hybrid models performed in light of radiology requirements. The goal of this dissertation was to fill this gap by determining how these models performed in light of these real-world settings, as well as how these settings impact the shape of a tumor that would be found in radiology presentations.

5. Development and Deployment of an Interactive Web Application :

A fully functional web interface using the Streamlit framework (app.py) was developed and hosted on the Streamlit Community Cloud, which allows for real-time brain tumor classification of uploaded images. The publicly available web interface showcases the successful implementation of the research model in a user-friendly fashion, thus increasing the impact of the project.

6. Academic Contribution:

This thesis not only added to the research done in the area of tumor classification based on the machine and deep learning techniques but also made it easier to recognize the positive and negative aspects of the hybrid models through the proposed framework. The outcomes of this dissertation will be of great assistance to the future studies, radiologists, and medical imaging educators.

6.4 Limitations :

Several limitations revolve around the scope and generalisability of this study:

• Constraints on the Dataset :

Realistic and well-regarded datasets, but those from public repositories suffer from exactly that. As a result of this, the datasets do not truly represent modern multi-scanner protocols, 3D volumes, or emerging tumor variants.

• Approach of Binary Classification :

The research carried out in this dissertation used a binary classification methodology, that is, images were classified as "tumor" or "healthy". Although this was sufficient to represent the problem space as required for the purpose of this dissertation, this methodology does not reflect the actual needs of any radiology department since most of them require the type of tumor, the grade of aggressiveness, and the contextualizing of tumor findings.

• Limited Model Exploration :

In the research presented in this dissertation, the ResNet18 + SVM classifier is selected because it has been proven to be reliable for tumor applications. However, the study did not extend the exploration to other machine learning algorithms such as EfficientNet, Vision Transformers, and autoencoders; therefore, the breadth of performance evaluation is limited.

- **No Real-time or Streaming Evaluation :**

This dissertation evaluated the model offline. Real-time CAD systems require processing streams of data in real time, measuring latency, and showing robustness to concept drift—none of which were evaluated in this dissertation.

- **Lack of Adversarial Testing :**

Tumors can be very subtle in their presentation and may, therefore, easily manipulate ML models. Adversarial testing of the model was beyond the scope of this dissertation, and hence it is not certain how resilient the model would be to inputs with manipulation.

While these limitations do not take away from the importance of the results, they are important parameters in interpreting the results and suggest potential avenues for further research. No datasets other than those freely available were used, with no human participants. This thus fell within institutional ethical research requirements.

6.5 Future Research :

There are several ways that future research may build upon the research presented within this dissertation:

- **Multi-Grade Tumor Detection :**

A set of models to detect and assess the aggressiveness level of particular cancer types (for instance, the difference between low-grade cancer, high-grade cancer, and glioma cancer) can make CAD systems more practically useful for radiology departments.

- **Real-Time and Adaptive CAD :**

The implementation of stream learning, concept drift detection, and incremental training would allow the CAD system to adapt to the dynamic changes of tumor manifestations and imaging conditions.

- **Hybrid Models :**

Exploration of these potential strengths of using vision transformers, 3D CNN, autoencoder, and ensemble learning on a combination of models may help to improve tumor detection of subtle/low-contrast tumors.

- **Adversarial Robustness Testing :**

Evaluating the model on adversarial samples could provide some insight into the robustness of the model to artifact-rich or low-quality scans.

- **Integration with PACS/RIS Tools :**

Future studies may include work towards creating an interface where CAD system output can be integrated with GE, Siemens, and other usual enterprise tools used for automated as well as human-centered processes.

- **Dataset Diversification :**

It is recommended that future work utilize new data sets or real-world hospital captures to improve the generalizability of the models, and the above directions would serve to improve the operational effectiveness and scientific integrity of future CAD systems.

6.6 Final Comments :

This research work has tried to validate the efficiency of hybrid machine learning and deep learning techniques for brain tumor detection and proved that ResNet18 + SVM has the capability to deliver outstanding performance results in a lab setup. By doing a complete review of existing literature, analyzing data, and conducting experiments, it has been shown in this research work that hybrid techniques possess a significant potential and limitation in the development of tumor classification techniques in medical images.

The hybrid learning method shows much more accuracy in comparison with conventional methods of radiomic detection. However, hybrid learning alone cannot be effectively utilized in medical imaging. To successfully apply hybrid methods in medical imaging, it requires robust data, dynamic modeling, organizational accountability, appropriate governance, and radiologists who take informed decisions.

Thus, tumor classification is not merely a technological problem. On the contrary, tumor classification is a process that heavily relies upon the partnership between humans and computers and the developments within the medical sector. The discoveries of the above dissertation will contribute to the dynamic front of machine learning technology-assisted medical imaging solutions.

These results are consistent with the aims set out in Chapter 1 and represent significant developments on the way to assessing the importance of hybrid machine learning and deep learning in brain tumor classification.

REFERENCES AND BIBLIOGRAPHY

Aamir, M., Rahman, Z., Dayo, Z. A., Abro, W. A., Uddin, M. I., Khan, I., ... & Hu, Z. (2022). A deep learning approach for brain tumor classification using MRI images. *Computers and Electrical Engineering*, 101, 108105.

<https://dl.acm.org/doi/10.1016/J.COMPELECENG.2022.108105>

Asiri, A. A., Soomro, T. A., Shah, A. A., Pogrebna, G., Irfan, M., & Alqahtani, S. (2024). Optimized brain tumor detection: a dual-module approach for mri image enhancement and tumor classification. *IEEE access*, 12, 42868-42887.

<https://ieeexplore.ieee.org/iel7/6287639/10380310/10476596.pdf>

Basthikodi, M., Chaithrashree, M., Ahamed Shafeeq, B. M., & Gurpur, A. P. (2024). Enhancing multiclass brain tumor diagnosis using SVM and innovative feature extraction techniques. *Scientific Reports*, 14(1), 26023.

<https://www.nature.com/articles/s41598-024-77243-7.pdf>

Mahesh, T. R., Geman, O., Margala, M., & Guduri, M. (2023). The stratified K-folds cross-validation and class-balancing methods with high-performance ensemble classifiers for breast cancer classification. *Healthcare Analytics*, 4, 100247.

<https://www.sciencedirect.com/science/article/pii/S2772442523001144>

Rasheed, Z., Ma, Y. K., Ullah, I., Ghadi, Y. Y., Khan, M. Z., Khan, M. A., ... & Shehata, A. M. (2023). Brain tumor classification from MRI using image enhancement and convolutional neural network techniques. *Brain Sciences*, 13(9), 1320. <https://www.mdpi.com/2076-3425/13/9/1320>

Raza, A., Ayub, H., Khan, J. A., Ahmad, I., S. Salama, A., Daradkeh, Y. I., ... & Hamam, H. (2022). A hybrid deep learning-based approach for brain tumor classification. *Electronics*, 11(7), 1146. <https://www.mdpi.com/2079-9292/11/7/1146>

Riahi, F., Kiani, P., Golabbakhsh, A., Khanezarrin, M., Abbaspour, M., Tabatabaei, S. A. D., ... & Mohammadi, H. (2024). Comparison of PET/CT and PET/MRI in central nervous system

tumors, a narrative review. International journal of physiology, pathophysiology and pharmacology, 16(4), 89. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11411248/pdf/ijppp0016-0089.pdf>

Xie, Y., Zaccagna, F., Rundo, L., Testa, C., Agati, R., Lodi, R., ... & Tonon, C. (2022). Convolutional neural network techniques for brain tumor classification (from 2015 to 2022): Review, challenges, and future perspectives. *Diagnostics*, 12(8), 1850. <https://www.mdpi.com/2075-4418/12/8/1850>

Nickparvar, M. Brain tumor MRI dataset . Kaggle. <https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

Murtoza, L. Brain tumor multimodal image CT and MRI . Kaggle. <https://www.kaggle.com/datasets/murtozalikhon/brain-tumor-multimodal-image-ct-and-mri>

McNatt, G. Multi-Modality Brain Tumor MRI & PET DICOM Series. Kaggle. <https://www.kaggle.com/datasets/grantmcnatt/mri-and-pet-dice-similarity-dataset>

Badža, M. M., & Barjaktarović, M. Č. (2020). Classification of brain tumors from MRI images using a convolutional neural network. *Applied Sciences*, 10(6), 1999. <https://www.mdpi.com/2076-3417/10/6/1999>

Louis, D. N., Perry, A., Wesseling, P., et al. (2021). The 2021 WHO Classification of Tumors of the Central Nervous System: a summary. *Neuro-Oncology*, 23(8), 1231–1251. <https://academic.oup.com/neuro-oncology/article/23/8/1231/6311214?login=false>

Swati, Z. N. K., Zhao, Q., Kabir, M., et al. (2019). Brain tumor classification for MR images using transfer learning and fine-tuning. *Computerized Medical Imaging and Graphics*, 75, 34–46. <https://www.sciencedirect.com/science/article/abs/pii/S0895611118305937?via%3Dihub>

APPENDICES

Appendix A – Python Code (written in Jupyter Notebook) Used in the Study :

A.1 — Step1_Dataset_Inspection_and_Preparation .ipynb

A.2 — Step2_Preprocessing_and_Datasets .ipynb

A.3 — Step3_Feature_Extraction .ipynb

A.4 — Step4_Model_Training .ipynb

A.5 — Step5_Final_Evaluation.ipynb

A.6 — CT_images. ipynb

A.7 — PET_images .ipynb

A.6 — app.py

Appendix B – Additional Experimental Results

B.1 — Detailed Test Performance Metrics :

Modality	Task	Accuracy	Precision (macro)	Recall (macro)	F1-Score (macro)	ROC-AUC (macro)
MRI	Binary	0.9977	0.9975	0.9979	0.9977	1.0000
MRI	Multiclass	0.9751	0.9792	0.9719	0.9755	0.9986
CT	Binary	0.9697	0.9698	0.9697	0.9697	0.9958
CT	Multiclass	0.9372	0.7727	0.7432	0.7559	0.9711
PET	Binary	1.0000	1.0000	1.0000	1.0000	1.0000
PET	Multiclass	1.0000	1.0000	1.0000	1.0000	1.0000

B.2 — Confusion Matrix Highlights (Selected Examples) :

- MRI Binary (Test set size: 442) Actual Healthy: Predicted Healthy = 200, Predicted Tumor = 0 Actual Tumor: Predicted Healthy = 1, Predicted Tumor = 241
- PET Multiclass (Test set size: 9) Perfect diagonal – all 9 samples correctly classified (5 Glioma, 2 Meningioma, 2 Pituitary)

B.3 — Label Distribution Across Modalities :

Modality	Class	Count	Percentage
MRI	Healthy	2,000	45.3%
MRI	Tumor	2,413	54.7%
CT	Healthy	2,300	49.8%
CT	Tumor	2,318	50.2%
PET	Healthy	28	34.6%
PET	Tumor	53	65.4%

B.4 — Feature Summary :

Feature Type	Dimension	Description
Handcrafted	37	GLCM (20), Hu moments (7), Histogram (10)
Deep (ResNet18)	512	Average pooled embeddings from penultimate layer

Appendix C – Hardware & Software Environment :

C.1 Hardware Specification :

CPU : Intel Core i5

RAM : 8 GB

OS : Windows 11.

C.2 Software Versions :

- Python : 3.13.3
- PyTorch : version 2.5.x
- Scikit-learn : Version 1.3+
- Pandas : Latest 2.x release
- NumPy : version 2.1.x
- Streamlit : 1.34.0
- Matplotlib & Seaborn : Latest stable versions
- Other Packages : Installed via pip and compatible with Python 3.13.3

Appendix D – Dataset Description :

Files/Folders Used :

Main project folder : " Brain_Tumor_Project ".

Dataset subfolders:

- mri / (Tumor/Glioma, Meningioma, Pituitary; Healthy)
- ct / (Tumor/Glioma, Meningioma, Pituitary; Healthy)

- pet / (Tumor/Glioma, Meningioma, Pituitary; Healthy)

Processed folders:

- splits/ (train/val/test for each modality)
- features/ (handcrafted and deep .npy files)
- results/ (CSV tables, PNG figures)

Total images processed:

- MRI : 4,413
- CT : 4,618
- PET : 81

Appendix E – Ethical Considerations :

1. Only publicly available datasets were used.
2. No personal, sensitive, or patient-identifiable data was collected, created, or processed.
3. No human participants were involved in this study.
4. All experiments were conducted offline on locally stored datasets.
5. The research complies with institutional ethical guidelines for non-human subject studies.

Appendix F – Project Folder Structure :

Brain_Tumor_Project

- Step1_Dataset_Inspection_and_Preparation.ipynb
- Step2_Preprocessing_and_Datasets.ipynb
- Step3_Feature_Extraction.ipynb
- Step4_Model_Training.ipynb
- Step5_Final_Evaluation.ipynb
- CT_images.ipynb
- PET_images.ipynb
- mri (folder with Healthy and Tumor subfolders)
- ct (folder with Healthy and Tumor subfolders)
- pet (folder with Healthy and Tumor subfolders)
- splits (folder with train/val/test for mri, ct, pet)
- features (folder with all .npy feature files)
- results (folder with CSV tables and PNG figures)
- models (folder for saved model files)
- class_names.json
- dataset_summary_step1.xlsx
- app. py
- dissertation.docx